

# The tetrahedral Horn problem and asymptotics of $U(n)$ $6j$ symbols

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## Abstract

For a compact reductive group  $G$ , Wigner  $6j$ -symbols are maps between multiplicity spaces which encode the associativity constraint  $(V_\alpha \otimes V_\beta) \otimes V_\delta \cong V_\alpha \otimes (V_\beta \otimes V_\delta)$  for unitary irreducible representations of  $G$ . For  $G = SU(2)$ , their asymptotic behavior for large weights was conjectured by Ponzano-Regge and established by Roberts using geometric quantization techniques.

In this paper, we address the problem of asymptotic behavior of  $6j$ -symbols for  $G = U(n)$ . We show that their decay rate is polynomial or exponential depending on the existence of a tetrahedron in the space of Hermitian  $n$  by  $n$  matrices with edges determined by the 6 arguments of the  $6j$ -symbol. Our proof is inspired by probabilistic techniques from quantum information theory.

As an application of our results, we obtain a new proof of the strongly subadditive entropy inequality for tetrahedra in the space of Hermitian matrices (the tetrahedral Horn problem).

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## 1 Introduction

Given a complex reductive group  $G$  with unitary irreducible representations (irreps)  $V_\lambda$  labelled by highest weights  $\lambda$  along with tensor-product multiplicity spaces,  $M_{\alpha,\beta}^\gamma := \text{Hom}_G(V_\gamma, V_\alpha \otimes V_\beta)$ , the  $6j$ -symbols are

maps between these multiplicity spaces which encode the associativity constraint  $(V_\alpha \otimes V_\beta) \otimes V_\gamma \cong V_\alpha \otimes (V_\beta \otimes V_\gamma)$  between a triple of irreps,  $V_\alpha, V_\beta$ , and  $V_\delta$  of  $G$ . Specifically, a  $6j$ -symbol depends on six irreps,  $V_\alpha, V_\beta, V_\gamma, V_\delta, V_\epsilon$ , and  $V_\phi$ , and is the linear map between multiplicity spaces of the type

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} : M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon \rightarrow M_{\alpha,\phi}^\epsilon \otimes M_{\beta,\delta}^\phi,$$

corresponding to the block-wise components of the canonical isomorphism from  $\bigotimes_\gamma M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon$  to  $\bigotimes_\phi M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon$ . Whenever the group  $G$  is multiplicity-free, in the sense that the dimensions of all multiplicity spaces,  $m_{\alpha,\beta}^\gamma := \dim M_{\alpha,\beta}^\gamma$ , are either zero or one, the corresponding  $6j$ -symbol is merely a scalar quantity.

**Wigner's  $6j$ -symbol for  $SU(2)$ :** The group  $SU(2)$  is multiplicity-free and the  $6j$ -symbols for  $SU(2)$  are, depending on the normalization and/or phase convention, are also referred to as Wigner  $6j$ -symbols or Wigner  $6j$ -coefficients [Wig93] or the Racah  $W$ -coefficients [Rac42]. The  $6j$ -symbols for  $SU(2)$  play a foundational role in the quantum theory of angular momentum, modelling the recoupling theory of three angular momenta, and thus they are used frequently in calculations involving the scattering of multiple particles in nuclear, atomic and molecular physics [VMK88; Edm96; Aqu+12]. Beyond the direct application to the theory of angular momenta, the  $6j$ -symbols for  $SU(2)$  also constitute a primitive object in the theory of spin networks [Pen71; BS10], representing a minimal non-trivial closed spin network. General spin networks are networks of trivalent vertices and connections labelled by irreps (spins) of  $SU(2)$  which provide a discrete/combinatorial and manifestly invariant model with applications to gauge theories on graphs or manifolds [Bae96], invariants of knots and manifolds [Res87; TV92], topological quantum field theory [DJN93], quantum computing [MR05], quantum change-point discrimination [AH11], and various aspects of proposed models for quantum gravity [Bae99; BC98; Bar03; Rov04].

Throughout this paper, by the *asymptotics* of  $6j$ -symbols, we are referring to their behaviour in the regime wherein the six highest weights,  $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$ , are becoming larger and larger in a suitable sense. The asymptotics of the  $6j$ -symbol for  $SU(2)$ , wherein the irreps of dimension  $\ell + 1$  may be identified either by integers  $\ell \in \{0, 1, \dots\}$  or half-integers  $j = \frac{\ell}{2} \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$ , were initially investigated by Wigner [Wig60, Chapter 27]. In the context of the quantum theory of angular momenta, the limit of large  $\ell$ , or equivalently large spins  $j$ , may also be interpreted as the limit wherein Planck's constant tends to zero,  $\hbar \propto \frac{1}{k} \rightarrow 0$ , which suppresses the effects of quantizing angular momentum; therefore, these asymptotic limits are also known as semiclassical limits. Given three particles with spins  $\frac{a}{2}$ ,  $\frac{b}{2}$  and  $\frac{d}{2}$  and total spin  $\frac{e}{2}$ , the  $6j$ -symbol,  $\left\{ \begin{matrix} a & b & c \\ d & e & f \end{matrix} \right\}$ , can be interpreted as the quantum mechanical amplitude to observe the first and second particles to have combined spin  $\frac{e}{2}$  given the second and third have combined spin  $\frac{f}{2}$  [Wig60; Rob99]. Wigner considered the probability for this event to occur in the asymptotic limit of large spins, i.e., the square of the  $6j$ -symbol, and conjectured the estimate:

$$\left\{ \begin{matrix} a & b & c \\ d & e & f \end{matrix} \right\}^2 \approx \frac{1}{3\pi V}$$

where  $V$  is the volume of the Euclidean tetrahedron with side-lengths  $(a, b, c, d, e, f)$ . Building upon the work of Wigner, Ponzano and Regge conjectured a formula for the amplitude of the  $6j$ -symbol itself in the limit of large spins [PR68]. Later, using techniques from geometric quantization theory, Roberts rigorously proved an asymptotic formula which is equivalent to the conjecture of Ponzano and Regge (up to minor differences) [Rob99]:

**Theorem 1** (Roberts). *If  $(a, b, c, d, e, f)$  are integer side-lengths of a Euclidean tetrahedron with volume  $V$  and corresponding exterior dihedral angles  $(\theta_a, \theta_b, \theta_c, \theta_d, \theta_e, \theta_f)$ , then*

$$\left\{ \begin{matrix} ka & kb & kc \\ kd & ke & kf \end{matrix} \right\} \stackrel{k \rightarrow \infty}{\sim} \sqrt{\frac{2}{3\pi V k^3}} \cos \left( \sum_\ell (k\ell + 1) \frac{\theta_\ell}{2} + \frac{\pi}{4} \right),$$

where the sum is over the six sides  $\ell$ . Otherwise, if there does not exist a Euclidean tetrahedron with side-lengths  $(a, b, c, d, e, f)$ , then  $\left\{ \begin{matrix} ka & kb & kc \\ kd & ke & kf \end{matrix} \right\}$  decays to zero at an exponential rate as  $k \rightarrow \infty$ .

It is worth emphasizing that Chen, Ismail, and Simeonov [CIS99] obtained a rigorous proof of the Ponzano and Regge asymptotics (with minor correction due to [Wat99]) slightly before Chen, Ismail, and Simeonov

using estimates from the theory of hypergeometric functions. Additional proofs of this asymptotic correspondence between  $6j$ -symbols for  $SU(2)$  and Euclidean tetrahedra have emerged in recent years including the elementary proof by Gurau which applies Stirling's approximation directly to Racah's single-sum formula [Gur08] as well as group integral approach to evaluate so-called simple spin networks as Feynman graphs [FK00; FL03]. See also the paper by Bartlett and Nzaganya for an accessible presentation of these kinds of calculations [BN23]. Analogous asymptotic formulas have also been obtained for the  $6j$  symbol for the quantum group  $U_q(\mathfrak{sl}_2)$  for a  $k$ -dependent parameter  $q$  [TW06]. The asymptotics calculated by Taylor and Woodward encode (i) the existence of a spherical tetrahedron on the unit sphere with prescribed edgelengths whenever  $q = \exp(i\pi/r(k))$  is an  $r(k)$ th root of unity or (ii) the existence of a hyperbolic tetrahedron with prescribed edgelengths when  $q = \exp(\pi/r(k))$  is positive real (here  $r(k) = k(r-2) + 2$  for  $r > 2$  a fixed integer) [TW06].

More generally, that the asymptotics of representation-theoretic quantities (such as  $6j$ -symbols) are controlled by the existence of some critical geometric object is explained by the theory of geometric quantization [Rob02]. Geometric quantization, broadly speaking, associates classical phase spaces modelled by symplectic manifolds to quantum phase spaces modelled by Hilbert spaces [Rob02; GS82; Kir04]. In particular, Kirillov's orbit principle [Kir04] proposes an association between (i) the coadjoint orbits of a compact Lie group  $G$  passing through an integral weight  $\lambda$  to (ii) the corresponding unitary irreps of  $G$  of highest weight  $\lambda$ . Furthermore, products of coadjoint orbits correspond to tensor products of irreps and moreover symplectic quotients of the resulting products of orbits are identified with invariant vectors in the resulting tensor products of irreps [Rob02; GS82]. For the aforementioned case of  $SU(2)$ , the coadjoint orbits can be identified the  $SO(3)$  rotations of vectors in Euclidean  $\mathbb{R}^3$  (i.e. spheres of fixed radius), and thus the asymptotics of representation-theoretic quantities for  $SU(2)$  are controlled by the existence of corresponding geometric objects in Euclidean  $\mathbb{R}^3$  with prescribed edge-lengths [Rob02].

**$6j$ -symbols for  $SU(n)$ :** The purpose of this paper is prove statements concerning the asymptotics of the  $6j$  symbol for the group  $U(n)$  (or equivalently for  $SU(n)$ ) wherein the polynomial irreps  $V_\lambda$  are labelled by highest-weights  $\lambda$  which can also be identified with Young diagrams. One of the primary difficulties in proving statements about  $6j$  symbols for  $SU(n)$  for  $n \geq 3$  is the presence of non-trivial multiplicity spaces, i.e. the dimensions of tensor-product multiplicity spaces for  $SU(n)$ , also known as Littlewood-Richardson coefficients, can be greater than one. Because of these difficulties, over the years, considerable attention has been placed on the study of  $SU(n)$  (or  $U_q(\mathfrak{sl}_n)$ )  $6j$  symbols which are multiplicity-free in the sense that the dimensions of each of the multiplicity spaces is either one or zero (in the latter case the  $6j$  symbol defined to be zero). For instance, tensor products involving symmetric representations, where  $\lambda = (k)$ , or conjugates of symmetric representations,  $\lambda = (k^{(n-1)})^1$ , are necessarily multiplicity free; Nawata, Pichai, and Zodinmawia conjectured closed form expressions for the  $U_q(\mathfrak{sl}_n)$   $6j$  symbols when  $a, b, d$  and  $e$  are integers and either  $(\alpha, \beta, \delta, \epsilon) = ((a), (b^{(n-1)}), (d), (e^{(n-1)}))$  and  $a + d = b + e$  (Type I) or  $(\alpha, \beta, \delta, \epsilon) = ((a), (b), (d^{(n-1)}), (e^{(n-1)}))$  and  $a + b = d + e$  (Type II) as well as the cases which can be related to these by known symmetries of the corresponding  $6j$  symbol [NPZ13]. Building upon this work, Alekseev, Morozov, and Sleptsov studied both the asymptotics of and symmetries in each of the Type I and Type II cases [AMS20]. More recently, Alcock-Zeilinger et al. have derived formulae for the family of  $SU(n)$   $6j$  symbols where at least one pair of non-adjacent edges, say  $(\alpha, \delta)$  correspond to a pair of fundamental/defining representations, i.e.,  $\alpha, \delta = (1)$  [Alc+23]. Shortly thereafter, Keppeler, Plätzer, and Sjö Dahl expanded upon these results to derive formulae for the family of  $SU(n)$   $6j$  symbols where either (i) at least one pair of non-adjacent edges consists of fundamental or adjoint representations, i.e.,  $\alpha, \delta \in \{(1), (2, 1^{(n-2)})\}$ , or (ii) at least one vertex, say  $(\alpha, \beta, \phi)$ , consists entirely of fundamental or adjoint representations, i.e.,  $\alpha, \beta, \phi \in \{(1), (2, 1^{(n-2)})\}$  [KPS24]. As stated by Coquereaux, “apart from very few special cases (low dimension of irreps and no multiplicity), ... we are not aware of any published general result.” [Coq23].

TC: Some residual research-notes concerning the  $6j$  symbols:

- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- Aquilanti et al. provides an excellent introduction to subject of classical  $6j$  symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]

<sup>1</sup>Here  $k^{(n-1)}$  is shorthand for a string  $k, k, \dots, k$  consisting of  $n-1$  repetitions of  $k$ .

- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum  $6j$  symbols for generic dimension  $n$  but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum  $6j$  symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum  $6j$  symbols for all  $n$  [LS23].

### Our contributions:

Regarding the tetrahedral Horn problem, we derive a new family of inequalities constraining the set of tetrahedral eigenvalues in terms of the so-called hook-rescaled Schur polynomials  $\varphi_\lambda(x) = s_\lambda(x)/H_\lambda$  where  $s_\lambda(x)$  is the Schur polynomial in the eigenvalues  $x = (x_1, \dots, x_n)$  and where  $H_\lambda$  is the product of the hook-lengths of all boxes in the Young diagram  $\lambda$ .

*thm:nec\_tet*

**Theorem 2.** *If  $(a, b, c, d, e, f)$  are the eigenvalues of positive semidefinite  $n \times n$  matrices  $(A, B, C, D, E, F)$  satisfying  $A+B=C$ ,  $B+D=F$ ,  $C+D=E$  and  $A+F=E$ , then for all triples of Young diagrams  $(\alpha, \beta, \delta)$ ,*

$$\varphi_\alpha(a)^{\frac{2}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{2}{3}} \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_\phi(f)^{\frac{1}{3}} \varphi_\gamma(c)^{\frac{1}{3}} \varphi_\epsilon(e)^{\frac{1}{3}}. \quad (1)$$

*eq:nec\_tet*

Moreover, we demonstrate the power of these inequalities by proving the following distance bound.

*thm:suff\_tet*

**Theorem 3.** *If  $(a, b, c, d, e, f)$  are non-negative eigenvalues satisfying the trace conditions and also Eq. (1) for all triples  $(\alpha, \beta, \delta)$  such that  $|\alpha| + |\beta| + |\delta| = k$  for some positive integer  $k$ , then*

$$\frac{1}{|e|} D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq O \left( n \sqrt{\frac{\log k}{k}} \right), \quad (2)$$

where  $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \geq 0$  with equality holding if and only if  $(a, b, c, d, e, f)$  are tetrahedral.

In other words, the satisfaction of Eq. (1) for Young diagrams of all sizes as  $k \rightarrow \infty$  is actually sufficient to conclude  $(a, b, c, d, e, f)$  are tetrahedral.

Along the way, we derive an analogous set of inequalities for the original Horn problem along with an accompanying distance bound.

*thm:horn\_orig\_poly\_ineqs*

**Theorem 4.** *Let  $(a, b, c)$  be non-negative eigenvalues satisfying  $|c| = |a| + |b|$ . If there exists positive semidefinite matrices  $A, B, C \geq 0$  with eigenvalues  $(a, b, c)$  satisfying  $A+B=C$ , then for all Young diagrams  $\gamma$ , we have*

$$\varphi_\gamma(c) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^{\gamma} > 0}} \varphi_\alpha(a) \varphi_\beta(b). \quad (3)$$

*eq:polynomial\_horn\_ineq*

Moreover, if Eq. (3) holds for all Young diagrams  $\gamma \vdash k$  of size  $k$ , then

$$\frac{1}{|c|} D(a, b, c) \leq O \left( n \sqrt{\frac{\log k}{k}} \right). \quad (4)$$

where  $D(a, b, c) := \min_{(A, B) \in \mathcal{O}_a \times \mathcal{O}_b} \|\text{eig}(A+B) - c\|_1$ .

Our approach bypasses the difficulties presented by multiplicities by focusing only on the *norm* (specifically the Hilbert-Schmidt, Frobenius or Schatten 2-norm) of the  $6j$  symbol for  $\text{SU}(n)$ :

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2 := \text{Tr} \left[ \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix}^\dagger \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right].$$

Following the procedure provided by geometric quantization, one expects the asymptotics of the  $6j$ -symbol for  $\text{U}(n)$  to be controlled by existence of some kind of tetrahedral object in space of coadjoint orbits for  $\text{U}(n)$ .

In particular, the coadjoint orbits of  $U(n)$  can be identified with the orbits of  $n \times n$  Hermitian matrices,  $X$ , under the conjugation action,  $X \mapsto U^{-1}XU$ , by unitary  $n \times n$  matrices  $U \in U(n)$ ; in other words, coadjoint orbits are associated to sets of  $n \times n$  Hermitian matrices sharing a common set of eigenvalues  $x = (x_1 \geq \dots \geq x_n)$ .

**Theorem 5** (Informal). *If  $(a, b, c, d, e, f)$  are the eigenvalues of  $n \times n$  of positive semidefinite matrices  $(A, B, C, D, E, F)$  satisfying*

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad \text{\textcolor{blue}{\{eq:tet_cond\}}} \quad (5)$$

*then there exists a sequence of Young diagrams  $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$  satisfying*

$$\frac{\alpha_k}{k} \rightarrow a, \quad \frac{\beta_k}{k} \rightarrow b, \quad \frac{\gamma_k}{k} \rightarrow c, \quad \frac{\delta_k}{k} \rightarrow d, \quad \frac{\epsilon_k}{k} \rightarrow e, \quad \frac{\phi_k}{k} \rightarrow f, \quad \text{\textcolor{blue}{\{eq:siz_convergence\}}} \quad (6)$$

*such that*

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2 \geq \frac{1}{\text{poly}(k)}. \quad (7)$$

*Otherwise, if  $(a, b, c, d, e, f)$  are not the eigenvalues of any matrices satisfying Eq. (5), then for any sequence of Young diagrams satisfying Eq. (6), we have exponential decay in the value of  $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2$ .*

Before continuing, it is worth emphasizing the similarities and differences between our result and the asymptotic formula proven by Roberts above [Rob99]. First and foremost, our result goes beyond that of Roberts in the sense that it applies to the higher-rank Lie group  $SU(n)$  for  $n \geq 3$ . Moreover, our result, even when restricted to the case for  $SU(2)$ , goes beyond the statement of proven by Roberts in the sense our result is applicable to *irrational* eigenvalues/edge-lengths, whereas the former is only applicable to *integer*, or with a minor re-scaling, *rational* eigenvalues/edge-lengths. Nevertheless, one drawback of our result is that it merely promises the *existence* of some sequence of Young diagrams,  $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ , converging to the eigenvalues  $(a, b, c, d, e, f)$ , whereas the result of Roberts provides a concrete statement about the particular sequence of *scaled* Young diagrams  $(k\alpha, k\beta, k\gamma, k\delta, k\epsilon, k\phi)$ .

Another drawback of our result is that it merely provides a polynomial lower bound on the value of  $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2$  and does not provide an asymptotic formula which would serve as the a proper generalization of the Ponzano and Regge formula above [PR68]. The Ponzano and Regge formula for  $SU(2)$  was traditionally obtained using the method of *stationary phases* which relies on both an integral formula for the  $6j$ -symbol with exponential integrand along with a geometric understanding of the *isolated* critical points for the exponential phase [PR68; Gur08]. In the case of  $SU(2)$  with generic edge-lengths, there is a formulation involving exactly *two* isolated critical points corresponding the  $SO(3)$ -orbit  $[\tau]$  of a Euclidean tetrahedron  $\tau$  with given edge-lengths and (ii) the  $SO(3)$ -orbit  $[\tau']$  of the Euclidean tetrahedron  $\tau'$  which is the mirror-image of the tetrahedron  $\tau$  with the same fixed side lengths. Generalizing this type of criticality analysis to  $SU(n)$  for  $n \geq 3$  appears to be a significant challenge, and even if an analysis was performed, one would still need to compute a corresponding Hessian for each critical point and evaluate the total contribution [Rob02]. Perhaps another approach, such as the approach adopted by Taylor and Woodward (see also [CIS99]) to prove asymptotics for the  $6j$ -symbol for  $U_q(\mathfrak{sl}_2)$  using recursion relations [TW06] (see also the formulas for the  $6j$ -symbols for  $gl_n$  in terms of generalized hypergeometric functions due to [Art24; Art23; Art24]).

**Eigenvalue estimation:** Our approach to proving the above theorem relies on the concept of *weak Schur sampling*, a probabilistic technique developed in the context of quantum information theory for estimating the eigenvalues of density matrices (trace one, positive semidefinite matrices), which model the state of quantum systems [KW01; Chr08; OW16; HM02; CHW07; Hay17]. **TODO: Finish the explanation of the eigenvalue-estimation idea once a suitable presentation of the all the “T” functions have been fixed.**

**The tetrahedral Horn problem:** The problem of determining the set of eigenvalues  $(a, b, c)$  of Hermitian matrices  $(A, B, C)$  satisfying  $A + B = C$  has been investigated for over a century [Wey12; Fan49; Lid53; Wie55; TF71; TF70] and has come to be known as *Horn’s problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities.

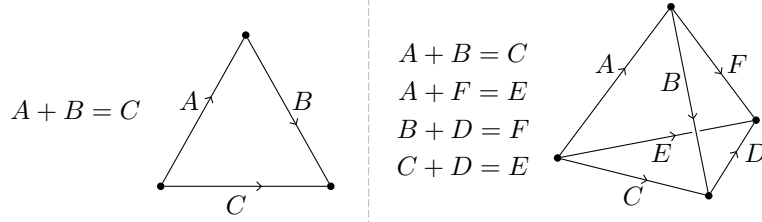


Figure 1: (Left) The *triangular* Horn's problem subjects three Hermitian matrices,  $(A, B, C)$ , to the constraint  $A + B = C$  forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn's problem subjects six Hermitian matrices,  $(A, B, C, D, E, F)$ , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices. [fig:tetrahedron\\_labelling\\_scheme](#)

In 1998, Klyachko provided a solution to Horn's problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn's original conjecture [Ful98]. These linear inequalities, now known as the *Horn inequalities* describe the convex polytope of admissible values for  $(a, b, c)$ . Recent developments on Horn's problem include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues  $\gamma$  of the sum  $C = A + B$  if  $A$  and  $B$  are sampled uniformly and independently from the respective sets of Hermitian matrices with eigenvalues  $a$  and  $b$  by Coquereaux, McSwiggen, and Zuber in series of papers [CZ18; CZ18; CMZ19; Zub18]. For a pedagogical and historical introduction to Horn's problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00] or alternatively [NP06, Appendix D]. See also the article by Daftuar and Hayden for an account of applications to topics in quantum information theory [DH05].

In parallel to Horn's problem, numerous variations and extensions of Horn's problem have been explored by various authors. For example, numerous authors have considered the admissible eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00; AMW01].

In this paper, we consider a generalization of Horn's problem investigated previously only briefly in [CŞW18]. While Horn's problem aims to characterize the eigenvalues  $(a, b, c)$  of a triple of  $n \times n$  Hermitian matrices  $(A, B, C)$  which are subjected to the constraint  $A + B = C$ , the generalization we consider here aims to characterize the eigenvalues  $(a, b, c, d, e, f)$  of a *sextuple* of  $n \times n$  matrices  $(A, B, C, D, E, F)$  subject to four constraints:  $A + B = C$ ,  $B + D = F$ ,  $D + C = E$  and  $A + F = E$ . These constraints arise naturally from building a tetrahedral arrangement of  $n \times n$  Hermitian matrices inside the vector space of such Hermitian matrices with edges labelled according to Fig. 1.

Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 1), in the special case of  $2 \times 2$  Hermitian matrices, this connection is much deeper. It is well-known that the space of admissible eigenvalues for a triangular arrangement of  $2 \times 2$  traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *triangles* in  $\mathbb{R}^3$  [HK96; FKL10; Knu00]. This correspondence generalizes to the tetrahedral case in the sense that the admissible eigenvalues for a tetrahedral arrangements of  $2 \times 2$  traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *tetrahedra* in  $\mathbb{R}^3$ . While this latter result is straightforward to prove and also well-known, indeed it forms the basis of the semiclassical asymptotics of the  $SU(2)$  Wigner  $6j$  symbols [Rob99], it leads to a complete solution to the tetrahedral Horn's problem for  $2 \times 2$  matrices; specifically, the moduli space of Euclidean tetrahedra in  $\mathbb{R}^3$  is provided by the positivity of so-called Cayley-Menger determinants which are polynomial inequalities in the side-lengths  $(a, b, c, d, e, f)$  [SJS18]. Even for  $n = 2$ , we remark that there is a strong qualitative difference between the tetrahedral Horn problem and the original Horn problem. In particular, while the Horn's problem is completely characterized by a *finite* number of *linear* inequalities in the eigenvalues (indeed the solution space is an instance of a moment polytope known as *Horn's polytope*), the solution space for the tetrahedral Horn's problem is not convex nor a polytope.

More recently, Alekseev et al. have investigated a few variants of the tetrahedral Horn problem concerning either the singular values of products of invertible matrices or the tropical singular values of concatenations of weighted planar networks [Ale+25]. These variants are termed *multiple* Horn problems in [Ale+25]; the tetrahedral Horn problem we consider here is termed the *additive* multiple Horn problem in the language of [Ale+25].

**Quantum marginals & the  $6j$ -symbol for the symmetric group  $S_k$ :** The techniques used in this paper to derive bounds on the  $6j$  symbol for the unitary group  $U(n)$  are closely related to the techniques used in an earlier paper by Christandl, Şahinoğlu, and Walter which derived analogous bounds on the  $6j$  symbol for the symmetric group  $S_k$  [CSW18]. In that paper, the authors considered the problem of determining whether there exists a density matrix  $\rho_{ABC}$  with partial traces  $\rho_{AB}, \rho_{BC}, \rho_A, \rho_B$ , and  $\rho_C$  with a given sextuple of eigenvalues [CSW18]. Such problems arise naturally in the context of quantum information theory and are typically referred to as quantum marginals problems [CSW18; Fra22]. In particular, they proved the  $6j$  symbols for the symmetric group  $S_k$  (referred to as the recoupling coefficients) in the limit as  $k \rightarrow \infty$  either decay exponentially or can be bounded from below by a polynomial factor depending on whether there exists a density matrix with the prescribed marginal spectra [CSW18]. Moreover, they also demonstrated how the tetrahedral Horn problem may be viewed as a special case of the quantum marginals problem and, in a complementary manner, furthermore demonstrated how, by restricting to the subgroup  $S_k \subset U(k)$  of permutation matrices and applying Nishiyama’s restriction theorem [Nis00], the  $6j$  symbols for unitary group  $U(n)$  may be viewed as a special case of the  $6j$  symbols for the symmetric group  $S_k$  [CSW18]. This latter result obtained by [CSW18] is a generalizes how the Littlewood-Richardson coefficients can be cast as a special case of the Kronecker coefficients [Lit58; Mur55; IP23]. The approach taken by this paper differs from that of [CSW18] in that we prove the correspondence between solutions to the tetrahedral Horn problem on one hand and the asymptotics of the  $6j$  symbol for  $U(n)$  on the other hand *directly*, rather than first pass through the quantum marginals problem or through the  $6j$  symbols for the symmetric group. We note that this more direct approach was initiated, in the  $n = 2$  case, by Backens [Bac10].

**Structure of the paper:** TODO: Descide on the structure of the paper? Finally, we prove an upper bound on general  $6j$  symbols using the Cauchy-Schwarz inequality which, in the  $n \rightarrow \infty$  asymptotic limit, gives rise to a novel proof of a known strongly subadditive inequality involving unnormalized entropies.

## 2 Tetrahedral eigenvalues

sec:tet\_eig

### 2.1 The tetrahedral Horn problem

prob:tri\_problem

**Definition 6** (Trigonal Horn problem). A triple  $(a, b, c) \in \text{Eig}(n)^{\times 3}$  of candidate eigenvalues are said to be *trigonal* if there exists a corresponding triple  $(A, B, C)$  of  $d \times d$  Hermitian matrices satisfying (i) the linear constraint

$$A + B = C, \quad \text{\texttt{\{eq:tri_linear_constraints\}}} \quad (8)$$

and (ii) the three eigenvalue constraints

$$\text{eig}(A) = a, \quad \text{eig}(B) = b, \quad \text{eig}(C) = c. \quad \text{\texttt{\{eq:tri_eigenvalue_constraints\}}} \quad (9)$$

The problem of determining the set of trigonal eigenvalues  $(a, b, c)$  is known as *Horn problem*.

prob:tet\_problem

**Definition 7** (Tetrahedral Horn problem). A sextuple  $(a, b, c, d, e, f) \in \text{Eig}(n)^{\times 6}$  of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple  $(A, B, C, D, E, F)$  of  $n \times n$  Hermitian matrices satisfying (i) the four linear constraints

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad \text{\texttt{\{eq:tet_linear_constraints\}}} \quad (10)$$

and (ii) the six eigenvalue constraints

$$\text{eig}(A) = a, \quad \text{eig}(B) = b, \quad \text{eig}(C) = c, \quad \text{eig}(D) = d, \quad \text{eig}(E) = e, \quad \text{eig}(F) = f. \quad \text{\texttt{\{eq:tet_eigenvalue_constraints\}}} \quad (11)$$

The problem of determining the set of tetrahedral eigenvalues  $(a, b, c, d, e, f)$  will be referred to as the *tetrahedral Horn problem*.

The tetrahedral Horn problem can be viewed as a generalization of the Horn's problem in a variety of ways. For instance, a triplet  $(a, b, c)$  of eigenvalues is trigonal if and only if the sextuple  $(a, b, c, 0, c, b)$  is tetrahedral. More generally, it is possible to view the tetrahedral Horn problem as the conjunction of four separate instances of Horn's original problem subject to additional matching constraints. Specifically, a sextuple of eigenvalues  $(a, b, c, d, e, f)$  is tetrahedral only if each of the following four triplets are trigonal:

$$(a, b, c), \quad (b, d, f), \quad (c, d, e), \quad (a, f, e). \quad \text{\texttt{\{eq:four_faces\}}} \quad (12)$$

If each of the above triplets are trigonal, then there exists a set of twelve Hermitian matrices with the corresponding eigenvalues satisfying

$$A + B = C, \quad \tilde{B} + D = F, \quad \tilde{C} + \tilde{D} = E, \quad \tilde{A} + \tilde{F} = \tilde{E}. \quad \text{\texttt{\{eq:four_faces_matrices\}}} \quad (13)$$

The tetrahedral Horn problem goes beyond four separate instances of Horn's original problem in that it additionally requires the primed and unprimed Hermitian matrices to coincide:<sup>2</sup>

$$A = \tilde{A}, \quad B = \tilde{B}, \quad C = \tilde{C}, \quad D = \tilde{D}, \quad E = \tilde{E}, \quad F = \tilde{F}. \quad (14)$$

### 2.2 Deviation Measures

In order to quantify the degree to which a candidate collection of eigenvalues is compatible, either in the case of Horn's original problem or in the the case of the tetrahedral Horn problem, it will be helpful to define some measures of deviation or distance which are non-negative and vanish if and only if the eigenvalues are compatible.

<sup>2</sup>Note that by conjugating the third and fourth conditions in Eq. (13) by unitaries  $U$  and  $V$  respectively, it is possible to set  $\tilde{C} \mapsto U^{-1}\tilde{C}U = C$  and  $\tilde{F} \mapsto V^{-1}\tilde{F}V = F$ , and thus one could assume without loss of generality that  $C = \tilde{C}$  and  $F = \tilde{F}$ .

**Proposition 8.** Let  $1 \leq p \leq \infty$ . Then for pairs of Hermitian matrices  $X, Y \in \text{Herm}(n)$  with eigenvalues  $\text{eig}(X) = x = (x_1 \geq \dots \geq x_n)$  and  $\text{eig}(Y) = y = (y_1 \geq \dots \geq y_n)$ , one has

$$\|X - Y\|_p \geq \|x - y\|_p. \quad (15)$$

Moreover, if  $S \subseteq \text{Herm}(n)$  is a unitarily invariant and closed subset of Hermitian matrices. Then

$$\min_{Y \in S} \|X - Y\|_p = \min_{y \in \text{eig}(S)} \|x - y\|_p, \quad (16)$$

In terms of Horn's original problem, we can consider the set  $S$  of Hermitian matrices which are of the form  $A + B$  for some  $(A, B) \in \mathcal{O}_a \times \mathcal{O}_b$ .

**Proposition 9.** For a given triple of eigenvalues,  $(a, b, c)$ , and  $1 \leq p \leq \infty$ , let

$$D_p(a, b, c) := \min_{U, V \in \text{U}(n)} \|\text{eig}(U \cdot [a] + V \cdot [b]) - c\|_p. \quad (17)$$

Then  $D_p(a, b, c) \geq 0$  with equality holding if and only if  $(a, b, c)$  are trigonal. Moreover,

$$D_p(a, b, c) = \min_{A, B, C \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_c} \|A + B - C\|_p. \quad (18)$$

For the tetrahedral Horn problem, there is some additional freedom in how one incorporates the various constraints. The measure of deviation which emerges naturally from our analysis is the following.

**Definition 10.** For a given sextuple of eigenvalues,  $(a, b, c, d, e, f)$ , and  $1 \leq p \leq \infty$ , let

$$D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} := \min_{U, V, W \in \text{U}(n)} \left( \|\text{eig}(U \cdot [a] + V \cdot [b]) - c\|_1^2 + \|\text{eig}(V \cdot [b] + W \cdot [d]) - f\|_1^2 + \|\text{eig}(U \cdot [a] + V \cdot [b] + W \cdot [d]) - e\|_1^2 \right)^{\frac{1}{2}}. \quad (19)$$

Then  $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \geq 0$  with equality holding if and only if  $(a, b, c, d, e, f)$  are trigonal.

## 2.3 Low dimensional solutions

**Remark 11** (Traces are tetrahedral). In the original version of Horn's problem, the eigenvalues  $(a, b, c)$  of Hermitian matrices  $(A, B, C)$  satisfying  $A + B = C$  necessarily must satisfy a basic equality condition known as the *trace condition*:

$$|a| + |b| = |c|, \quad \text{\texttt{\{eq:tri\_trace\_cond\}}} \quad (20)$$

where  $|x| := \sum_i x_i$  is shorthand for the sum of the eigenvalues  $x = (x_1 \geq \dots \geq x_n)$ . This condition must hold because  $A + B = C$  implies  $\text{Tr}(C) = \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$  and  $\text{Tr}(X) = |x|$  for any Hermitian matrix  $X$  with eigenvalues  $x$ . For the tetrahedral Horn's problem, a set of tetrahedral eigenvalues  $(a, b, c, d, e, f)$  necessarily satisfy four *trace conditions*:

$$|a| + |b| = |c|, \quad |b| + |d| = |f|, \quad |d| + |c| = |e|, \quad |a| + |f| = |e|. \quad \text{\texttt{\{eq:tet\_trace\_conds\}}} \quad (21)$$

Furthermore, for the trivial case where  $n = 1$ , the above trace constraints are both necessary and sufficient for the tetrahedral Horn problem.

\texttt{exam:d\\_2}

**Example 12** (Tetrahedral Horn's problem for  $d = 2$ ). Here we articulate the solution to the tetrahedral Horn's problem for  $2 \times 2$  Hermitian matrices. We restrict our focus to the case of *traceless*  $2 \times 2$  Hermitian matrices<sup>3</sup>. In this context, the sextuple of eigenvalues  $(a, b, c, d, e, f)$  can be identified with a sextuple of non-negative real numbers  $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$  such that the sextuple of traceless eigenvalues are

$$(a, -a), \quad (b, -b), \quad (c, -c), \quad (d, -d), \quad (e, -e), \quad (f, -f). \quad \text{\texttt{\{eq:traceless\_two\_dim\_tetra\}}} \quad (22)$$

<sup>3</sup>In Section 2.4, Corollary 17, we will see that this restriction can be made without loss of generality.

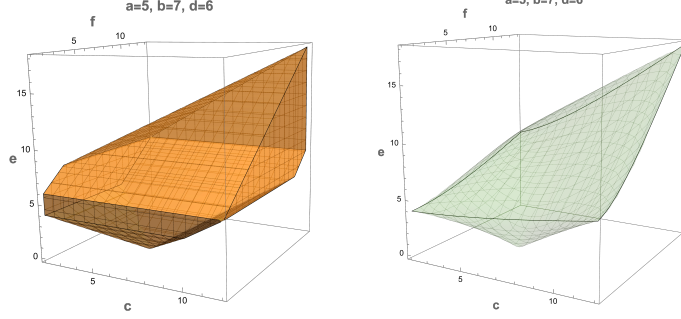


Figure 2: A slice of the set of admissible side-lengths  $(a, b, c, d, e, f)$  for Euclidean tetrahedron along the coordinates  $(a, b, d) = (5, 7, 6)$  (Left) The polyhedron of values  $(c, e, f)$  satisfying the triangle inequalities for each face Eq. (25). (Right) The set of all values  $(c, e, f)$  satisfying both Eq. (25) and Eq. (26). {eq:tetrahedron(26)}{eq:li\_space\_example}

Moreover, each traceless  $2 \times 2$  Hermitian matrix,  $X$ , may be identified with a three-dimensional vector  $(x_1, x_2, x_3) \in \mathbb{R}^3$  using the standard Pauli-matrix basis  $\{\sigma_1, \sigma_2, \sigma_3\}$  according to the formula:

$$X = x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (23)$$

Under this identification, the eigenvalues of  $X$  are determined by the Euclidean length of the vector  $(x_1, x_2, x_3)$ :

$$\text{eig}(X) = \text{eig} \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (24)$$

Furthermore, if  $A$  is determined by the coordinates  $(a_1, a_2, a_3)$  and  $B$  by the coordinates  $(b_1, b_2, b_3)$ , then their sum  $C = A + B$  is determined by the coordinates  $(c_1, c_2, c_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$ . Consequently, in this setting, the tetrahedral Horn's problem reduces to problem of describing the space of all possible edge-lengths  $(a, b, c, d, e, f)$  of a Euclidean tetrahedron in  $\mathbb{R}^3$ . This problem admits of a well-studied solution [Ric02; WD09], namely, the conjunction of a quadruplet of triangle-inequalities

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (25) \quad \text{ {eq:quadruplet_triangle_constraints} }$$

(which ensure the possibility of forming a Euclidean triangle for each of the four faces), together with a single non-linear inequality of the form

$$\text{CM}(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & c^2 & e^2 \\ 1 & a^2 & 0 & b^2 & f^2 \\ 1 & c^2 & b^2 & 0 & d^2 \\ 1 & e^2 & f^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad (26) \quad \text{ {eq:volume_positive_constraint} }$$

The quantity  $\text{CM}(a, b, c, d, e, f)$  is known as a Cayley-Menger determinant and, when  $\text{CM}(a, b, c, d, e, f)$  is positive and when the triangle inequalities above are also satisfied,  $\text{CM}(a, b, c, d, e, f)$  is always equal to *square* of the volume of the corresponding tetrahedron [SJS18, Equation 5.2]<sup>4</sup>. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (25) and Eq. (26).

<sup>4</sup>The triangle inequalities  $|a - b| \leq c \leq a + b$  are also equivalent to the non-negativity of the *area*,  $\text{CM}(a, b, c)$ , of the triangle with side-lengths  $a, b$  and  $c$  where  $\text{CM}(a, b, c)^2 = -\frac{1}{16} \det \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & a^2 \\ 1 & a^2 & 0 \end{pmatrix}$  is equivalent to Heron's formula.

## 2.4 Symmetries of tetrahedral eigenvalues

sec:symmetries

There are numerous types of symmetries present in the statement of Definition 7. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. These symmetries can, in many cases, be used to simplify the problem by selecting a canonical representative of the corresponding equivalence class; in other cases, it will be useful to leave Definition 7 as stated.

prop:scaling

**Proposition 13** (Scaling symmetry). *Let  $(a, b, c, d, e, f)$  be a sextuple of candidate eigenvalues and let  $s \in (0, \infty)$  be a positive real number. Then the scaled eigenvalues,  $(sa, sb, sc, sd, se, sf)$ , are tetrahedral if and only if the original, unscaled eigenvalues  $(a, b, c, d, e, f)$  are tetrahedral.*

*Proof.* The statement follows from the fact that  $\text{eig}(sX) = s \text{eig}(X)$  holds for all Hermitian matrices  $X$  and positive scalars  $s$  and that each of the constraints in Eq. (10) are homogeneous in the six Hermitian matrices  $(A, B, C, D, E, F)$ ; for example, we have  $sA + sB - sC = 0$  if and only if  $s(A + B - C) = 0$  if and only if  $A + B - C = 0$ .  $\square$

One may wonder whether it is possible to extend the scaling symmetry to consider scaling by negative  $s < 0$ . Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrices both negates and reverses the order of its eigenvalues.

prop:inversion

**Proposition 14** (Inversion symmetry). *Let  $(a, b, c, d, e, f)$  be a sextuple of candidate eigenvalues. Given any list of non-increasing eigenvalues  $x = (x_1 \geq \dots \geq x_n)$ , let  $x^*$  denote the inverted list of resorted and negated eigenvalues:  $x^* = (-x_d \geq \dots \geq -x_1)$ . Then  $(a, b, c, d, e, f)$  are tetrahedral if and only if  $(a^*, b^*, c^*, d^*, e^*, f^*)$  are tetrahedral.*

*Proof.* The statement holds because (i)  $(A, B, C, D, E, F)$  satisfy Eq. (10) if and only if  $(A', B', C', D', E', F') := (-A, -B, -C, -D, -E, -F)$  satisfy Eq. (10), and (ii)  $\text{eig}(X) = x$  if and only if  $\text{eig}(-X) = x^*$ .  $\square$

The inversion symmetry Proposition 14 is just one element belonging to a group of 48 symmetries consisting of inversions of eigenvalues,  $x \rightarrow x^*$ , or permutations of elements in the sextuple  $(a, b, c, d, e, f)$ .

prop:tet\_symmetry

**Proposition 15** (Oriented Tetrahedral symmetry). *The sextuple  $(a, b, c, d, e, f)$  of eigenvalues are tetrahedral if and only if any (and thus all) of the following sextuples of eigenvalues are tetrahedral:*

$$(a \ f \ e \ d^* \ c \ b) \quad (c \ e^* \ d^* \ f \ b \ a^*) \quad \text{eq:gens_of_tet_sym} \quad (27)$$

Moreover, these two operations generate a finite group of order 48 whose action, grouped by conjugacy classes, produces:

$$\begin{array}{cccc} (a \ b \ c \ d \ e \ f) & (a^* \ c \ b \ d \ f \ e) & (a \ c^* \ b^* \ d^* \ f^* \ e^*) & (a^* \ b^* \ c^* \ d^* \ e^* \ f^*) \\ (d \ b \ f \ a \ e \ c) & (b^* \ a^* \ c^* \ e \ d \ f) & (b \ a \ c \ e^* \ d^* \ f^*) & (d^* \ b^* \ f^* \ a^* \ e^* \ c^*) \\ (d^* \ e \ c \ a^* \ b \ f) & (f^* \ b \ d^* \ c^* \ e^* \ a^*) & (f \ b^* \ d \ c \ e \ a) & (d \ e^* \ c^* \ a \ b^* \ f^*) \\ (a \ e^* \ f^* \ d \ b^* \ c^*) & (e \ d^* \ c \ b^* \ a \ f^*) & (e^* \ d \ c^* \ b \ a^* \ f) & (a^* \ e \ f \ d^* \ b \ c) \\ & (c \ b^* \ a \ f \ e \ d) & (c^* \ b \ a^* \ f^* \ e^* \ d^*) & \\ & (a \ f \ e \ d^* \ c \ b) & (a^* \ f^* \ e^* \ d \ c^* \ b^*) & \\ (b \ f^* \ d^* \ e \ c \ a) & (b \ d \ f \ e^* \ a^* \ c^*) & (b^* \ d^* \ f^* \ e \ a \ c) & (b^* \ f \ d \ e^* \ c^* \ a^*) \\ (c \ a^* \ b \ f^* \ d^* \ e^*) & (d \ f^* \ b^* \ a^* \ c^* \ e^*) & (d^* \ f \ b \ a \ c \ e) & (c^* \ a \ b^* \ f \ d \ e) \\ (e \ c^* \ d \ b \ f \ a^*) & (f \ e^* \ a^* \ c \ b \ d^*) & (f^* \ e \ a \ c^* \ b^* \ d) & (e^* \ c \ d^* \ b^* \ f^* \ a) \\ (f \ a \ e \ c^* \ d \ b^*) & (e \ a \ f^* \ b \ d^* \ c) & (e^* \ a \ f \ b^* \ d \ c^*) & (f^* \ a^* \ e^* \ c \ d^* \ b) \\ (f^* \ d \ b^* \ c \ a \ e) & (d^* \ c^* \ e^* \ a \ f^* \ b^*) & (d \ c \ e \ a^* \ f \ b) & (f \ d^* \ b \ c^* \ a^* \ e^*) \\ (e^* \ f \ a^* \ b^* \ c^* \ d) & (c^* \ e \ d \ f^* \ b^* \ a) & (c \ e^* \ d^* \ f \ b \ a^*) & (e \ f^* \ a \ b \ c \ d^*) \\ (c^* \ d^* \ e^* \ f \ a^* \ b) & & & (c \ d \ e \ f^* \ a \ b^*) \\ (b^* \ c \ a \ e^* \ f^* \ d^*) & & & (b \ c^* \ a^* \ e \ f \ d) \end{array} \quad \text{eq:all_of_tet_sym} \quad (28)$$

*Proof.* To begin, notice that each of the operations on  $(a, b, c, d, e, f)$  appearing in Eq. (27) are invertible (indeed the first operation has order two, while the second operation has order four). The claim that these two operations generate a finite group of order 48 can be checked explicitly using computer software. Now suppose  $(a, b, c, d, e, f)$  are the eigenvalues of a sextuple of Hermitian matrices  $(A, B, C, D, E, F)$  satisfying

Eq. (10). Then, for example, the sextuple  $(A', B', C', D', E', F') = (A, F, E, -D, C, B)$  (which has the eigenvalues  $(a, f, e, d^*, c, b)$ ) is also tetrahedral because

$$\begin{aligned} A' + B' - C' &= (A + F - E) = 0, \\ B' + D' - F' &= -(B + D - F) = 0, \\ C' + D' - E' &= -(C + D - E) = 0, \\ A' + F' - E' &= (A + B - C) = 0. \end{aligned} \tag{29}$$

Moreover, as the operation mapping  $(a, b, c, d, e, f)$  to  $(a^*, c, b, d, f, e)$  is invertible, one also has that if  $(a, b, c, d, e, f)$  is not tetrahedral, then neither is  $(a^*, c, b, d, f, e)$ . An analogous argument holds for the operation mapping  $(a, b, c, d, e, f)$  to  $(c, e^*, d^*, f, b, a^*)$  with  $(A', B', C', D', E', F')$  given by  $(C, -E, -D, F, B, -A)$ . □

**Proposition 16** (Shifting symmetry). *Let  $(a, b, c, d, e, f)$  be a sextuple of candidate eigenvalues and let  $x, y, z \in \mathbb{R}$  be fixed. Given any list of non-increasing eigenvalues  $x = (x_1 \geq \dots \geq x_n)$ , let  $x + x := (x_1 + x \geq \dots \geq x_d + x)$ . Then define shifted eigenvalues by the formula*

$$\begin{aligned} \tilde{a} &:= a + x, & \tilde{b} &:= b + y, & \tilde{c} &:= c + x + y, \\ \tilde{d} &:= d + z, & \tilde{e} &:= e + x + y + z, & \tilde{f} &:= f + y + z. \end{aligned} \tag{30}$$

The shifted eigenvalues  $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$  are tetrahedral if and only if the unshifted eigenvalues  $(a, b, c, d, e, f)$  are tetrahedral.

*Proof.* Given a Hermitian matrix  $X$  with eigenvalues  $\text{eig}(X) = x$  and a constant  $w \in \mathbb{R}$ , the eigenvalues of  $X + wI$  (where  $I$  is the identity matrix) are

$$\text{eig}(X + wI) = x + w = (x_1 + w \geq \dots \geq x_d + w). \tag{31}$$

The statement then follows because

$$(A + xI, B + yI, C + (x + y)I, D + zI, E + (x + y + z)I, F + (y + z)I) \tag{32}$$

satisfy the tetrahedral constraints in Eq. (10) if and only if  $(A, B, C, D, E, F)$  satisfy the tetrahedral constraints in Eq. (10) □

As a corollary of the shifting symmetries (Proposition 16) one concludes that to solve the tetrahedral Horn's problem, in a certain sense, it suffices to solve the tetrahedral Horn's problem merely for traceless matrices (Corollary 17) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 18).

**Corollary 17** (Traceless eigenvalues). *Let  $(a, b, c, d, e, f)$  be a sextuple of candidate eigenvalues and for any list of  $n$  eigenvalues  $x = (x_1 \geq \dots \geq x_n)$ , let  $x^0 := x - |x|/n$  denote the traceless list of eigenvalues obtained by shifting  $x$  by  $-|x|/n$ . Then  $(a, b, c, d, e, f)$  are tetrahedral if and only if (i) their traces  $(|a|, |b|, |c|, |d|, |e|, |f|)$  are tetrahedral (they satisfy Eq. (21)) and (ii)  $(a^0, b^0, c^0, d^0, e^0, f^0)$  are tetrahedral.*

*Proof.* This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (21)) and applications of the shifting symmetry Proposition 16 with  $(x, y, z)$  values

$$x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D). \tag{33}$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated  $(x, y, z)$  values. □

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn's problem it suffices to solve the tetrahedral Horn's problem for positive definite spectra only. Indeed, one can simply take the shifting parameters  $(x, y, z)$  appearing in Proposition 16 to be sufficiently large. The following corollary establishes, in a certain sense, the minimum shifting required to ensure strictly positive eigenvalues for all six Hermitian matrices. Note that in subsequent sections we will often explicitly assume the candidate eigenvalues  $(a, b, c, d, e, f)$  are either non-negative or positive for the sake of technical convenience or sometimes to establish a connection with density matrices from quantum theory (which are positive semidefinite matrices with unit trace).

**Corollary 18** (Positive eigenvalues). *If  $(a, b, c, d, e, f)$  are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$c_d \geq a_d + b_d, \quad f_d \geq b_d + d_d, \quad e_d \geq d_d + c_d, \quad e_d \geq a_d + f_d. \quad (34)$$

Moreover, if  $(a, b, c, d, e, f)$  are a sextuple of candidate eigenvalues satisfying Eq. (34), then for any  $(x, y, z) \in \mathbb{R}^3$  satisfying

$$x > -a_d, \quad y > -b_d, \quad z > -d_d, \quad (35)$$

the resulting shifted eigenvalues  $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$  (given by Eq. (30)) are all positive.

*Proof.* That tetrahedral eigenvalues satisfy Eq. (34) follows from four applications of the well-known inequality  $c_d \geq a_d + b_d$  for Horn's problem where  $C = A + B$  [Bha01]. As a reminder, the smallest eigenvalue  $x_n$  of a matrix  $X$  is characterized by a minimization:  $x_d = \min_{\|v\|=1} \langle v, Xv \rangle$  which produces

$$c_d = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = a_d + b_d. \quad (36)$$

Four applications of this inequality produces Eq. (34). Now if  $(x, y, z)$  satisfy Eq. (35), we have for all  $j$

$$a_j + x \geq a_d + x > 0, \quad (37)$$

meaning  $a + x = (a_1 + x \geq \dots \geq x_d + x)$  is strictly positive. Similar statements hold for  $b, c, d, e$  and  $f$ ; for instance, for all  $j$ ,

$$e_j + x + y + z \geq e_d + x + y + z > e_d - a_d - b_d - d_d \geq 0, \quad (38)$$

meaning  $e + x + y + z$  is strictly positive also.  $\square$

**Remark 19.** Although this section listed a handful of symmetries of the tetrahedral Horn's problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case  $2 \times 2$  Hermitian matrices, an additional class of symmetries on the set of edge-lengths  $(a, b, c, e, d, f)$  of tetrahedra, known as the *Regge symmetries*, hold [AI19; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths  $(a, b, c, d, e, f)$ , then there also exists a tetrahedron with edge lengths

$$(s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}. \quad (39)$$

Combining this symmetry with the permutational symmetries given by Proposition 15 (and noting that for the eigenvalues  $x$  of traceless  $2 \times 2$  Hermitian matrices,  $x = (x, -x)$ , we have  $x^* = x$ ) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

$$\begin{aligned} (a, b, c, d, e, f) & \quad (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ (a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) & \quad (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ (s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) & \quad (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{aligned} \quad (40)$$

where

$$s_1 = \frac{a + b + d + e}{2}, \quad s_2 = \frac{a + c + d + f}{2}, \quad s_3 = \frac{b + c + e + f}{2}. \quad (41)$$

**Remark 20.** The Regge symmetry of a tetrahedron can be understood as a consequence of an earlier discovery of a suprising symmetry of Wigner  $6j$  symbol for  $SU(2)$  originally due to Regge [Reg59]. The paper due to Boalch provides a comprehensive and modern treatment of the Regge symmetry for the  $6j$  symbol for  $SU(2)$  by viewing it as a consequence of a simpler symmetry of the  $6j$ -symbol for  $SU(3)$  [Boa07]. That the Regge symmetry of a tetrahedron emerges from the Regge symmetry of the associated  $6j$  symbols can be viewed as a consequence of the semiclassical limit of the latter [PR68; Rob99]. It has also been shown that the Regge symmetry of Euclidean tetrahedra is actually a scissors-congruence preserving both the volume and Dehn invariants [AI19].

At the moment, it remains unclear if there exists analogous symmetries for the generalized Wigner  $6j$  symbol for  $n \geq 3$  (although there are conjectured symmetries for the  $6j$  symbols for  $U_q(\mathfrak{sl}_n)$  which would follow from the eigenvalue hypothesis [MS18; LS23]). It could also be possible for additional symmetries to be present in the tetrahedral Horn problem for  $n \geq 3$  which are *not* present in the  $6j$  symbol for  $SU(n)$ .

### 3 New inequalities for tetrahedral eigenvalues

The eigenvalues,  $x = (x_1, \dots, x_n)$ , of an  $n \times n$  Hermitian matrix  $X$  determine which  $U(n)$  orbit,

#### 3.1 Schur-Weyl duality

The purpose of this section is to review the representation theory of  $U(n)$ , or equivalently, the polynomial representation theory of  $GL(n)$ .

will be to examine the Schur-Weyl decomposition of the  $k$ -th tensor power  $A^{\otimes k}$ . To begin, we recall the Schur-Weyl decomposition of  $(\mathbb{C}^n)^{\otimes k}$  as a representation of the group  $S_k \times U(n)$ .

**Proposition 21** (Schur-Weyl duality). *Let  $k$  and  $n$  be positive integers. Then the decomposition of  $(\mathbb{C}^n)^{\otimes k}$  into irreducible subrepresentations of  $S_k \times U(n)$ , known as the Schur-Weyl decomposition, is given by*

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda \vdash_n k} W_\lambda \otimes V_\lambda^n \quad \text{\texttt{eq:schur_weyl_decomp}} \quad (42)$$

where the direct summands are indexed by Young diagrams  $\lambda$  with  $k$  boxes and at most  $n$  rows, and where  $W_\lambda$  supports the irreducible representation of  $S_k$  and  $V_\lambda^n$  supports the irreducible representation of  $U(n)$ .

The dimension of the  $V_\lambda^n$  is given by Weyl's dimension formula [Hal08; IN66],

$$\dim(V_\lambda^n) = \frac{\Delta_n(\tilde{\lambda})}{\Delta_n(\rho)}, \quad \text{\texttt{eq:weyl_dim}} \quad (43)$$

whereas the dimension of  $W_\lambda$  may be given by [GW00, Corollary 9.1.5],

$$\dim(W_\lambda) = \frac{|\lambda|!}{\tilde{\lambda}_1! \tilde{\lambda}_2! \cdots \tilde{\lambda}_n!} \Delta_n(\tilde{\lambda}) \quad \text{\texttt{eq:frob_dim}} \quad (44)$$

where  $\tilde{\lambda} = \lambda + \rho$ , where  $\rho = (n-1, n-2, \dots, 1, 0)$  is known as the Weyl vector and where  $\Delta_n(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j)$  is the Vandermonde determinant. Note that the representation  $W_\lambda$  of  $S_k$  is independent of  $n$ ; indeed, the right hand side of Eq. (44) does not depend on  $n$  and coincides with the Hook-length formula [GW00, Corollary 9.1.6] (any value of  $n$  which is at least the depth of  $\lambda$  suffices).

**Definition 22.** Given a non-zero positive semidefinite  $n \times n$  matrix  $A$ , a positive integer  $x$ , and a Young diagram  $\alpha \vdash_n x$ , let

$$\tau_\alpha(A) := \text{Tr}[\underline{A}^{\otimes x} P_\alpha], \quad (45)$$

where  $\underline{A} = \frac{A}{\text{Tr}(A)}$  is the normalization of  $A$  and  $P_\alpha$  projects onto the isotypic component of  $(\mathbb{C}^n)^{\otimes x}$  indexed by  $\alpha$ .

As  $P_\alpha$  is  $U(n)$ -equivariant, one observes that  $\tau_\alpha(A)$  depends only on the eigenvalues of  $A$ . The following result, known as spectrum or eigenvalue estimation, states that the map  $\alpha \mapsto \tau_\alpha(A)$  for  $A$  positive semidefinite is concentrated those Young diagrams  $\alpha$  whose shape matches the shape of the eigenvalues  $a = \text{eig}(A)$  [ARS88; KW01; HM02].

\text{\texttt{prop:eig\_est}}

**Proposition 23** (Eigenvalue estimation). *Let  $A$  be a positive semidefinite matrix with eigenvalues  $a = \text{eig}(A)$  and let  $\alpha \vdash_n a$ . Then*

$$\tau_\alpha(A) \leq \dim(V_\alpha^n) \exp(-aD(\underline{\alpha}||\underline{a})), \quad (46)$$

where  $D(\underline{\alpha}||\underline{a})$  denotes the Kullback-Leibler divergence of  $\underline{\alpha}$  from  $\underline{a}$ .

*Proof.* See [KW01] for a proof (noting that  $\text{Tr}(\underline{A}) = 1$ ).

\text{\texttt{prop:completeness\_rel\_uni}} □

**Proposition 24.** *Let  $A$  be positive semidefinite  $n \times n$  matrix with  $\text{Tr}(A) = 1$  with eigenvalues  $a = \text{eig}(A)$  and rank  $r(a) = \text{rank}(A)$ . Then for any positive integer  $a$ ,*

$$\sum_{\alpha \vdash_{r(a)} a} \tau_\alpha(A) = 1, \quad (47)$$

where the sum is over Young diagrams  $\alpha$  with size  $a$  with depth at most  $r(a)$ .

**Definition 25.** Let  $(\pi_\lambda, V_\lambda^n)$  be an irreducible representation of  $U(n)$  labelled by the Young diagram  $\lambda$ . Given an  $n \times n$  matrix  $X$ , viewed as a linear operator on the defining representation  $V_{(1)}^n \simeq \mathbb{C}^n$ , we define  $\pi_\lambda(X)$  by the formula:

$$\pi_\lambda(X) := \frac{\text{Tr}_{W_\lambda}[\iota_\lambda^\dagger X^{\otimes |\lambda|} \iota_\lambda]}{\dim(W_\lambda)}$$

where  $\iota_\lambda : W_\lambda \otimes V_\lambda^n \rightarrow V_{(1)}^{\otimes |\lambda|}$  is the equivariant isometry for the  $\lambda$ -isotypic subspace of  $V_{(1)}^{\otimes |\lambda|}$ . This definition ensures that  $X^{\otimes k} \simeq \bigoplus_{\lambda \vdash k} \pi_\lambda(X)^{\oplus W_\lambda}$  holds for any positive integer  $k$ . When  $X$  is a unitary,  $\pi_\lambda(X)$  is the representation of  $X$  on  $V_\lambda^n$ , so this abuse of notation is warranted. If  $X \geq 0$  is positive semidefinite, then  $\pi_\lambda(X) \geq 0$  is positive semidefinite also. If  $\lambda = (1)$ , then  $\pi_{(1)}(X) = X$  as  $\dim(W_{(1)}) = 1$ .

Note also that the matrix coefficients of  $\pi_\lambda(X)$  are homogeneous degree  $|\lambda|$  polynomials in the matrix coefficients of  $X$  meaning for any scalar  $s \in \mathbb{C}$ ,

$$\pi_\lambda(sX) = s^{|\lambda|} \pi_\lambda(X). \quad (48)$$

Additionally, if  $X, Y$  are two matrices, we have

$$\pi_\lambda(XY) = \pi_\lambda(X)\pi_\lambda(Y). \quad (49)$$

The *character* of  $X$  satisfies

$$\chi_\lambda(X) := \text{Tr}(\pi_\lambda(X)) = \frac{\text{Tr}[P_\lambda X^{\otimes |\lambda|}]}{\dim(W_\lambda)} = s_\lambda(\text{eig}(X)),$$

where  $s_\lambda(\text{eig}(X))$  is the corresponding Schur-polynomial evaluated on the eigenvalues of  $X$ . For fixed positive integer  $k$ , we also define the *Plancherel distribution*  $\{\mu_\lambda | \lambda \vdash k\}$  over Young diagrams of size  $k$  according to

$$\mu_\lambda := \frac{(\dim W_\lambda)^2}{|\lambda|!}. \quad (50)$$

Moreover, we define the rescaling of  $\pi_\lambda(X)$  to be

$$\tilde{\pi}_\lambda(X) := \dim(W_\lambda) \pi_\lambda(X) = \text{Tr}_{W_\lambda}[\iota_\lambda^\dagger X^{\otimes |\lambda|} \iota_\lambda],$$

and the corresponding rescaled character is

$$\tilde{\chi}_\lambda(X) := \text{Tr}[\tilde{\pi}_\lambda(X)] = \text{Tr}[P_\lambda X^{\otimes |\lambda|}] = \dim(W_\lambda) s_\lambda(\text{eig}(X)),$$

where  $P_\lambda := \iota_\lambda^\dagger \iota_\lambda$  is the orthogonal projection operator onto the  $\lambda$ -isotypic subspace.

These rescaled characters emerge as coefficients in the following decomposition associated to an integral over Haar measure over  $U(n)$ ,

$$\int_{U(n)} dU (UXU^{-1})^{\otimes k} = \sum_{\lambda \vdash k} \tilde{\chi}_\lambda(X) \frac{P_\lambda}{\text{Tr}(P_\lambda)}. \quad (51)$$

This normalization scheme means that  $\tilde{\pi}_\lambda(X)$  is *not* a representation, while the character  $\tilde{\chi}_\lambda(X)$  satisfies

$$\sum_{\lambda \vdash k} \tilde{\chi}_\lambda(X) = \sum_{\lambda \vdash k} \text{Tr}[X^{\otimes k} P_\lambda] = \text{Tr}[X^{\otimes k}] = \text{Tr}[X]^k. \quad (52)$$

Therefore, whenever  $X$  is positive semidefinite and trace one (i.e., when  $X$  is a density matrix), we have that the map  $\lambda \mapsto \tilde{\chi}_\lambda(X)$  is a probability distribution when restricted to the set of Young diagrams with a fixed number  $k$  of boxes:

$$X \geq 0 \implies \tilde{\chi}_\lambda(X) \geq 0, \quad \text{and} \quad \text{Tr}(X) = 1 \implies \sum_{\lambda \vdash k} \tilde{\chi}_\lambda(X) = 1. \quad (53)$$

As the (rescaled) character  $\chi_\lambda(X)$  depends only on the eigenvalues  $\text{eig}(X)$  of  $X$ , we have

$$\chi_\lambda(X) = \chi_\lambda(\text{diag}(\text{eig}(X))). \quad (54)$$

Given a tuple of non-negative sorted eigenvalues  $x = (x_1 \geq \dots \geq x_n \geq 0)$  summing to one,  $\sum_{i=1}^n x_i = 1$ , the distribution  $\lambda \mapsto \tilde{\chi}_\lambda(\text{diag}(x))$  over  $\lambda \vdash k$  is denoted  $\text{SW}^k(x)$  and called the *Schur-Weyl* distribution in [OW16].

### 3.2 The binomial theorem

[sec:binomial\\_theorem](#)

The purpose of this section is to state and prove the binomial theorem, which serves as the foundation for all subsequent results in this paper. The binomial theorem expresses any polynomial irreducible representation of a matrix sum  $X + Y$  as a sum over the polynomial irreducible representations of the summands  $X$  and  $Y$ .

The theorem inherits its name from the non-commutative binomial expansion of the  $k$ -th tensor power of the sum  $X + Y$ ,

$$(X + Y)^{\otimes k} = \sum_{\ell=0}^k \binom{k}{\ell} \left[ \frac{1}{k!} \sum_{\sigma \in S_k} \sigma(X^{\otimes \ell} \otimes Y^{\otimes(k-\ell)}) \sigma^{-1} \right] \quad (55)$$

where  $\sigma \in S_k$  acts on  $(\mathbb{C}^n)^{\otimes k}$  by permutation of the  $k$  tensor factors. For an example, consider the case where  $k = 3$ :

$$\begin{aligned} (X + Y)^{\otimes 3} = & \underbrace{X \otimes X \otimes X}_{\ell=3} + \underbrace{X \otimes X \otimes Y + X \otimes Y \otimes X + Y \otimes X \otimes X}_{\ell=2} + \\ & + \underbrace{X \otimes Y \otimes Y + Y \otimes X \otimes Y + Y \otimes Y \otimes X}_{\ell=1} + \underbrace{Y \otimes Y \otimes Y}_{\ell=0}. \end{aligned} \quad (56)$$

As we have seen previously, the  $k$ -th tensor power of a matrix can also be decomposed into a sum over polynomial irreducible representations labelled by Young diagrams  $\gamma$  with sizes  $|\gamma| = k$ . Combining this formula with the non-commutative binomial expansion yields

$$(X + Y)^{\otimes k} = \sum_{\gamma \vdash k} \iota_{\gamma}(\pi_{\gamma}(X + Y) \otimes \text{id}_{W_{\gamma}}) \iota_{\gamma}^{\dagger} \quad (57)$$

where

$$\pi_{\gamma}(X + Y) = \frac{1}{\dim(W_{\gamma})} \sum_{\ell=0}^k \binom{k}{\ell} \text{Tr}_{W_{\gamma}}[\iota_{\gamma}^{\dagger}(X^{\otimes \ell} \otimes Y^{\otimes(k-\ell)}) \iota_{\gamma}], \quad (58)$$

where we have eliminated the symmetrization operation over  $\sigma \in S_k$  because the partial trace over  $W_{\gamma}$  is  $S_k$ -invariant. At this stage, for each summand indexed by  $0 \leq \ell \leq k$ , we can decompose  $X^{\otimes \ell}$  into representations of  $X$  labelled by  $\alpha$  with sizes  $|\alpha| = \ell$  and  $Y^{\otimes \ell}$  into representations of  $Y$  labelled by  $\beta$  with sizes  $|\beta| = k - \ell$ .

$$X^{\otimes \ell} = \sum_{\alpha \vdash \ell} \iota_{\alpha}(\pi_{\alpha}(X) \otimes \text{id}_{W_{\alpha}}) \iota_{\alpha}^{\dagger}, \quad Y^{\otimes k-\ell} = \sum_{\beta \vdash k-\ell} \iota_{\beta}(\pi_{\beta}(Y) \otimes \text{id}_{W_{\beta}}) \iota_{\beta}^{\dagger}. \quad (59)$$

Upon doing so, one encounters the composite operator  $\iota_{\gamma}^{\dagger}(\iota_{\alpha} \otimes \iota_{\beta})$  which maps from  $(V_{\alpha}^n \otimes W_{\alpha}) \otimes (V_{\beta}^n \otimes W_{\beta})$  to  $(V_{\gamma}^n \otimes W_{\gamma})$  in an  $S_{\ell} \times S_{k-\ell}$ -equivariant and  $U(n)$ -equivariant manner. By Schur's lemma and Schur-Weyl duality, it can be written as

$$\iota_{\gamma}^{\dagger}(\iota_{\alpha} \otimes \iota_{\beta}) = (\text{id}_{V_{\gamma}^n} \otimes \omega_{\alpha\beta}^{\gamma})(\iota_{\alpha\beta}^{\gamma} \otimes \text{id}_{W_{\alpha} \otimes W_{\beta}})^{\dagger} \quad (60)$$

where  $\iota_{\alpha\beta}^{\gamma} : C_{\alpha\beta}^{\gamma} \otimes V_{\gamma}^n \rightarrow V_{\alpha}^n \otimes V_{\beta}^n$  is the  $U(n)$ -equivariant isometry for the  $\gamma$ -isotypic subspace of  $V_{\alpha}^n \otimes V_{\beta}^n$  and  $\omega_{\alpha\beta}^{\gamma} : C_{\alpha\beta}^{\gamma} \otimes W_{\alpha} \otimes W_{\beta} \rightarrow W_{\gamma}$  the  $S_{\ell} \times S_{k-\ell}$ -equivariant isometry for the  $(\alpha, \beta)$ -isotypic subspace of  $W_{\gamma}$ .<sup>5</sup> Combining everything together, and rearranging we obtain the formula

$$\frac{\dim W_{\gamma} \pi_{\gamma}(X + Y)}{|\gamma|!} = \sum_{\alpha, \beta} \text{Tr}_{C_{\alpha, \beta}^{\gamma}} \left[ (\iota_{\alpha, \beta}^{\gamma})^{\dagger} \frac{\dim W_{\alpha} \pi_{\alpha}(X)}{|\alpha|!} \otimes \frac{\dim W_{\beta} \pi_{\beta}(Y)}{|\beta|!} (\iota_{\alpha, \beta}^{\gamma}) \right], \quad (61)$$

where we leave the summation over all pairs  $(\alpha, \beta)$  such that  $|\alpha| = \ell$ ,  $|\beta| = k - \ell$  for  $|\gamma| = k$  implicit simply because the multiplicity space  $C_{\alpha\beta}^{\gamma}$  is empty if  $|\alpha| + |\beta| \neq |\gamma|$  and thus the corresponding summand is taken to be zero.

In the remaining of the paper, we will use the above formula, Eq. (61), to prove numerous results concerning the eigenvalues of Hermitian matrices and their sums. In order to simplify the presentation of our results considerably, we introduce the following definitions.

<sup>5</sup>Schur-Weyl duality provides us with a isometric and bijective identification between the multiplicity spaces for  $V_{\gamma}^n$  inside  $V_{\alpha}^n \otimes V_{\beta}^n$  with respect to  $U(n)$  and  $W_{\alpha} \otimes W_{\beta}$  inside  $W_{\gamma}$  with respect to  $S_{\ell} \times S_{k-\ell}$ , hence why we use, by abuse of notation, the symbol  $C_{\alpha\beta}^{\gamma}$  in both cases.

**Definition 26** (Hook-rescaled representations). Given an  $n \times n$  matrix  $X$  and a Young diagram  $\lambda$ , define an operator  $\Phi_\lambda(X)$  acting on the representation space  $V_\lambda^n$  as any of the following equivalent expressions:

$$\Phi_\lambda(X) := \frac{\dim W_\lambda \pi_\lambda(X)}{|\lambda|!} = \frac{\pi_\lambda(X)}{H_\lambda} = \frac{\text{Tr}_{W_\lambda}[\iota_\lambda X^{\otimes |\lambda|} \iota_\lambda^\dagger]}{|\lambda|!}, \quad (62)$$

where  $H_\lambda = \prod_{b \in \lambda} h_\lambda(b)$  is the product of the Hook-lengths of the boxes in the Young diagram  $\lambda$ . Furthermore, let the trace of  $\Phi_\lambda(X)$  be denoted by

$$\varphi_\lambda(X) = \text{Tr}[\Phi_\lambda(X)] = \frac{\dim W_\lambda \chi_\lambda(X)}{|\lambda|!} = \frac{\chi_\lambda(X)}{H_\lambda} = \frac{\text{Tr}[P_\lambda X^{\otimes |\lambda|}]}{|\lambda|!}. \quad (63)$$

**Remark 27.** The operator  $\Phi_\lambda(X)$  and trace  $\varphi_\lambda(X)$  are simply rescalings of the representation matrix  $\pi_\lambda(X)$  of  $X$  and the character  $\chi_\lambda(X)$  of  $X$  respectively. This normalization scheme is such that summing the rescaled character over all Young diagrams  $\lambda$  with fixed size  $k$  or over all sizes yields respectively

$$\sum_{\lambda \vdash k} \varphi_\lambda(X) = \frac{\text{Tr}(X)^k}{k!}, \quad \sum_{\lambda} \varphi_\lambda(X) = \exp(\text{Tr}(X)). \quad (64)$$

Before stating the main result of this section, we first generalize the quantities  $\Phi_\lambda(X)$  and  $\varphi_\lambda(X)$  to consider arguments involving more than one matrix.

**Definition 28** (Mixed hook-rescaled representations). Given a pair  $(X, Y)$  of  $n \times n$  matrices and a triple of Young diagrams  $(\alpha, \beta, \gamma)$ , define an operator  $\Phi_\gamma^{\alpha\beta}(X, Y)$  acting on the representation space  $V_\gamma^n$  as

$$\Phi_\gamma^{\alpha\beta}(X, Y) := \text{Tr}_{C_{\alpha,\beta}^\gamma}[\iota_{\alpha,\beta}^{\gamma;\dagger}(\Phi_\alpha(X) \otimes \Phi_\beta(Y))\iota_{\alpha,\beta}^\gamma], \quad (65)$$

where  $\iota_{\alpha,\beta}^\gamma : C_{\alpha,\beta}^\gamma \otimes V_\gamma^n \rightarrow V_\alpha^n \otimes V_\beta^n$  is the  $U(n)$ -equivariant isometry for the  $\gamma$ -isotypic subspace of  $V_\alpha^n \otimes V_\beta^n$ . Note the sum over Young diagrams  $\alpha$ , and  $\beta$  is implicitly restricted to those satisfying  $|\alpha| + |\beta| = |\gamma|$ , since otherwise the multiplicity space  $C_{\alpha,\beta}^\gamma$  is zero-dimensional and thus the summand is taken to be zero. Analogously, let the trace of  $\Phi_\gamma^{\alpha\beta}(X, Y)$  be denoted by

$$\varphi_\gamma^{\alpha\beta}(X, Y) := \text{Tr}[\Phi_\gamma^{\alpha\beta}(X, Y)] = \text{Tr}[P_{\alpha,\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))], \quad (66)$$

where  $P_{\alpha,\beta}^\gamma = \iota_{\alpha,\beta}^\gamma \iota_{\alpha,\beta}^{\gamma;\dagger}$  is the projection operator onto the  $\gamma$ -isotypic subspace of  $V_\alpha^n \otimes V_\beta^n$ .

**Remark 29.** While  $\Phi_\lambda(X)$  is a homogeneous polynomial in the matrix coefficients of  $X$  of degree  $|\lambda|$  in the sense that  $\Phi_\lambda(sX) = s^{|\lambda|}\Phi_\lambda(X)$  holds for any  $s \in \mathbb{R}$ ,  $\Phi_\gamma^{\alpha\beta}(X, Y)$  is a bihomogeneous polynomial in  $X$  and  $Y$  of degrees  $|\alpha|$  and  $|\beta|$  respectively in the sense that for any  $s, t \in \mathbb{R}$ ,

$$\Phi_\gamma^{\alpha\beta}(sX, tY) = s^{|\alpha|}t^{|\beta|}\Phi_\gamma^{\alpha\beta}(X, Y). \quad (67)$$

**Remark 30.** The quantity  $\Phi_\gamma^{\alpha\beta}(X, Y)$  generalizes  $\Phi_\lambda(X)$  in the sense that one can take the triple  $(\alpha, \beta, \gamma) = (\lambda, (), \lambda)$  (where  $()$  denotes the Young diagram with zero boxes), in which case  $V_{\emptyset}^n$  supports the trivial one-dimensional representation with  $\Phi_{\emptyset}(Y) = 1$  and because  $\dim(C_{\lambda,(\emptyset)}^\lambda) = 1$ ,

$$\forall Y : \quad \Phi_\lambda^{\lambda,(\emptyset)}(X, Y) = \Phi_\lambda(X). \quad (68)$$

With these definitions in place, we are in a position to state the main result of this section.

*thm:binomial*

**Theorem 31** (Binomial theorem). For all  $n \times n$  matrices  $X, Y$ , and all Young diagrams  $\gamma$ ,

$$\Phi_\gamma(X + Y) = \sum_{\alpha, \beta} \Phi_\gamma^{\alpha\beta}(X, Y), \quad (69)$$

where the sum is implicitly taken over all Young diagrams such that  $|\alpha| + |\beta| = |\gamma|$ .

*Proof.* See preceding discussion. □

cor:bisummation

**Corollary 32.** *Let  $X, Y$  be  $n \times n$  matrices. Then the following identities hold:*

$$\forall \alpha, \beta : \sum_{\gamma} \varphi_{\gamma}^{\alpha\beta}(X, Y) = \varphi_{\alpha}(X) \varphi_{\beta}(Y), \quad (70)$$

$$\forall \gamma : \sum_{\alpha, \beta} \varphi_{\gamma}^{\alpha\beta}(X, Y) = \varphi_{\gamma}(X + Y). \quad (71)$$

*Proof.* The first identity involving a summation over  $\alpha$  and  $\beta$  is simply the result of taking the trace of both sides of Theorem 31. For the second identity, involving a summation over  $\gamma$ , follows from the resolution of the identity  $\sum_{\gamma \vdash |\alpha| + |\beta|} P_{\alpha\beta}^{\gamma} = \text{id}_{V_{\alpha}^n \otimes V_{\beta}^n}$ :

$$\varphi_{\alpha}(X) \varphi_{\beta}(Y) = \text{Tr}[\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y)] = \sum_{\gamma} \text{Tr}[P_{\alpha\beta}^{\gamma}(\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y))] = \sum_{\gamma} \varphi_{\gamma}^{\alpha\beta}(X, Y), \quad (72)$$

where the last equality is just the definition of  $\varphi_{\gamma}^{\alpha\beta}(X, Y)$ . Once again, the condition that  $|\gamma| = |\alpha| + |\beta|$  is implicitly present. □

**Remark 33.** One might consider what happens to the operator  $\Phi_{\gamma}^{\alpha\beta}(X, Y)$  when one sums over  $\gamma$  instead of  $(\alpha, \beta)$ . However, an analogous identity concerning a sum over  $\gamma$  of the operator  $\Phi_{\gamma}^{\alpha\beta}(X, Y)$  does not immediately follow because the tensor product  $\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y)$  is not necessarily block diagonalizable into  $\gamma$ -isotypic subspaces in any obvious way. One exception occurs when  $X = Y$ , in which case,

$$\pi_{\alpha}(X) \otimes \pi_{\beta}(X) = \sum_{\gamma} \iota_{\alpha\beta}^{\gamma}(\pi_{\gamma}(X) \otimes \text{id}_{C_{\alpha\beta}^{\gamma}}) \iota_{\alpha\beta}^{\gamma;\dagger}, \quad (73)$$

which is the operator version of the character product formula  $\chi_{\alpha}(X) \chi_{\beta}(X) = \sum_{\gamma} c_{\alpha\beta}^{\gamma} \chi_{\gamma}(X)$ .

### 3.3 Results for random matrices

sec:random\_matrices

**Proposition 34.** *Let  $x = (x_1 \geq \dots \geq x_n)$  be a set of eigenvalues and let  $X \sim \mathcal{O}_x$  denote a random Hermitian matrix of the form  $X = U[x]U^{-1}$  where  $U \in \text{U}(n)$  is sampled uniformly with respect to the normalized Haar measure on  $\text{U}(n)$ . Then for any Young diagram  $\lambda$ ,*

$$\mathbb{E}_{X \sim \mathcal{O}_x} \pi_{\lambda}(X) = \frac{s_{\lambda}(x)}{\dim V_{\lambda}^n} \text{id}_{V_{\lambda}^n}, \quad (74)$$

or equivalently, in terms of  $\Phi_{\lambda}(X)$  and  $\varphi_{\lambda}(x)$ ,

$$\mathbb{E}_{X \sim \mathcal{O}_x} \Phi_{\lambda}(X) = \frac{\varphi_{\lambda}(x)}{\dim V_{\lambda}^n} \text{id}_{V_{\lambda}^n}, \quad (75)$$

cor:random\_sum

**Corollary 35.** *Let  $(x, y)$  be a pair of eigenvalues. Then*

$$\mathbb{E}_{X \sim \mathcal{O}_x} \mathbb{E}_{Y \sim \mathcal{O}_y} \frac{\varphi_{\gamma}^{\alpha\beta}(X, Y)}{\dim V_{\gamma}^n} = c_{\alpha\beta}^{\gamma} \frac{\varphi_{\alpha}(x)}{\dim V_{\alpha}^n} \frac{\varphi_{\beta}(y)}{\dim V_{\beta}^n}. \quad (76)$$

lem:random\_matrices\_tet

**Lemma 36.** *Let  $(x, y, z)$  be a triple of eigenvalues. Then*

$$\mathbb{E}_{X \sim \mathcal{O}_x} \mathbb{E}_{Y \sim \mathcal{O}_y} \mathbb{E}_{Z \sim \mathcal{O}_z} \frac{\varphi_{\gamma \in \phi}^{\alpha\beta\delta}(X, Y, Z)}{\dim V_{\epsilon}^n} = \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \frac{\varphi_{\alpha}(x)}{\dim V_{\alpha}^n} \frac{\varphi_{\beta}(y)}{\dim V_{\beta}^n} \frac{\varphi_{\delta}(z)}{\dim V_{\delta}^n}. \quad (77)$$

### 3.4 Inequalities for Horn's original problem

In Section 3.2 we introduced to the quantity  $\varphi_\gamma^{\alpha\beta}(X, Y)$  and saw in Corollary 32 that it satisfied two summation identities. Under the assumption that the matrices  $X$ , and  $Y$  are positive semidefinite, we can conclude that  $\varphi_\gamma^{\alpha\beta}(X, Y)$  is non-negative. This observation implies the following upper bounds on  $\varphi_\gamma^{\alpha\beta}(X, Y)$ . lem:XY\_marginal\_upper\_bounds

**Lemma 37.** *Let  $X, Y \geq 0$  be positive semidefinite  $n \times n$  matrices. For all Young diagrams  $(\alpha, \beta, \gamma)$ , we have the inequality*

$$\varphi_\gamma^{\alpha\beta}(X, Y) \leq \min\{\varphi_\alpha(X)\varphi_\beta(Y), \varphi_\gamma(X + Y)\}. \quad (78)$$

*Proof.* The claim follows directly from Corollary 32 provided  $\varphi_\gamma^{\alpha\beta}(X, Y) \geq 0$ . However, this follows because  $\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}[P_{\alpha\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))]$  and  $P_{\alpha\beta}^\gamma \geq 0$  (it is a projection operator) and  $\Phi_\alpha(X), \Phi_\beta(Y) \geq 0$  because  $X, Y \geq 0$ . cor:orig\_horn\_ineq  $\square$

**Corollary 38.** *Let  $X, Y \geq 0$  be positive semidefinite  $n \times n$  matrices. Then the following inequalities hold:*

$$\forall \alpha, \beta: \quad \varphi_\alpha(X)\varphi_\beta(Y) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\gamma(X + Y), \quad (79)$$

$$\forall \gamma: \quad \varphi_\gamma(X + Y) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\alpha(X)\varphi_\beta(Y). \quad (80)$$

*Proof.* The claim follows by applying upper bounds from Lemma 37 to the summations in Corollary 32 being careful to note that if  $c_{\alpha\beta}^\gamma = 0$ , then  $\varphi_\gamma^{\alpha\beta}(X, Y) = 0$  and one only needs to consider the cases where  $c_{\alpha\beta}^\gamma \neq 0$ .  $\square$

### 3.5 The $6j$ symbol for $U(n)$

sec:sixj\_tet

The purpose of this section is to introduce a representation theoretic object, namely the  $6j$  symbol, and then in the subsequent sections show how it places constraints on whether a given sextuple of eigenvalues  $(a, b, c, d, e, f)$  is tetrahedral in a manner which generalizes how the Littlewood-Richardson coefficient encodes whether a given triple  $(a, b, c)$  is trigonal. For the case where  $n = 2$ , this correspondence is well-known [Rob99; PR68; Bac10], while for the case where  $n \geq 3$ , this correspondence was already established indirectly by Christandl, Şahinoğlu, and Walter [CSW18]. Here we establish this correspondence directly.

As we discussed in the introduction, the  $6j$  symbol emerges naturally when considering successive decompositions of the triple tensor product  $V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n$ . defn:sixj

**Definition 39.** Let  $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$  be a sextuple of Young diagrams with sizes satisfying

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\gamma| + |\delta| = |\epsilon|, \quad |\alpha| + |\phi| = |\epsilon|. \quad (81)$$

The  $U(n)$ - $6j$  operator is the operator<sup>6</sup>

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} : C_{\alpha, \beta}^\gamma \otimes C_{\delta, \gamma}^\epsilon \rightarrow C_{\alpha, \phi}^\epsilon \otimes C_{\beta, \delta}^\phi, \quad (82)$$

defined by

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} := \frac{1}{\dim(V_\epsilon)} \text{Tr}_{V_\epsilon} [\iota_{\alpha\phi}^{\epsilon; \dagger} (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^{\phi; \dagger}) (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) \iota_{\gamma\delta}^\epsilon], \quad (83)$$

{eq:sixj}

and depicted diagrammatically by

To illustrate the situation more concretely, consider the following three projection operators on the triple tensor product  $V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n$ :

$$P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}, \quad P_{\alpha\beta\delta}^\epsilon, \quad \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi. \quad (85)$$

In general,  $P_{\alpha\beta}^\gamma$  and  $P_{\beta\delta}^\phi$  do not necessarily commute with each other in the sense that for some values of  $\alpha, \beta, \delta, \gamma$ , and  $\phi$ , one observes

$$[P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}, \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi] \neq 0. \quad (86)$$

Nevertheless, the projection operators  $P_{\alpha\beta}^\gamma$  and  $P_{\beta\delta}^\phi$  each respectively commute with  $P_{\alpha\beta\delta}^\epsilon$  in the sense that for all  $\alpha, \beta, \gamma, \delta, \epsilon$ , and  $\phi$ ,

$$[P_{\alpha\beta\delta}^\epsilon, P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}] = 0, \quad [P_{\alpha\beta\delta}^\epsilon, \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi] = 0. \quad (87)$$

Let  $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$  and  $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$  be projection operators given by

$$\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} := P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi), \quad \Pi_{\alpha\beta\delta}^{\epsilon;\phi} := P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi). \quad (88)$$

[{eq:double\\_projs}](#)  
[prop:6j\\_norm\\_double\\_proj](#)

**Proposition 40.** For any sextuple of Young diagrams  $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ ,

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty = \left\| \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \right\|_\infty, \quad \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 = \frac{1}{\dim V_\epsilon^n} \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi}]. \quad (89)$$

### 3.6 Inequalities for the tetrahedral Horn problem

**Definition 41.** Let  $X, Y, Z$  be  $n \times n$  matrices. For any sextuple  $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$  of Young diagrams, let

$$\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) := \text{Tr}[(P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi) \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)]. \quad (90)$$

Remark that in our definition of  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ , we permit the sizes of the involved Young diagrams to be arbitrary. However, there are numerous conditions on the sizes of the Young diagrams  $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$  which are necessary (but not always sufficient) for  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$  to be non-zero; specifically,

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\alpha| + |\phi| = |\epsilon|, \quad |\gamma| + |\delta| = |\epsilon|. \quad (91)$$

<sup>6</sup>It is important to note that there does not seem to exist a universal convention concerning how 6j symbols are normalized; Coquereaux discusses this problem in [\[Coq23\]](#).

If we additionally let  $\Omega_{\alpha\beta\delta}^{X,Y,Z} := \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)$ , then we can compactly express  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$  as

$$\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}]. \quad (\text{eq:varphi_compact_tet}) \quad (92)$$

Unlike the quantity  $\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}[P_\alpha^\gamma \Phi_\alpha(X) \otimes \Phi_\beta(Y)]$  involving two matrices  $X, Y$  (see Section 3.2), the quantity defined here,  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ , is not necessarily nonnegative or even real valued even under the assumption that  $X, Y, Z \geq 0$  are positive semidefinite.

Nevertheless, summing over either  $\gamma$  or  $\phi$  restores non-negativity because it eliminates one of the potentially non-commuting projection operators  $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$  or  $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$ . We denote these partial sums of  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$  over  $\gamma$  and  $\phi$  respectively as

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) := \sum_\gamma \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}] = \text{Tr}[P_{\gamma\delta}^\epsilon (\Phi_\gamma^{\alpha\beta}(X, Y) \otimes \Phi_\delta(Z))], \quad (93)$$

$$\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) := \sum_\phi \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Omega_{\alpha\beta\delta}^{X,Y,Z}] = \text{Tr}[P_{\alpha\phi}^\epsilon (\Phi_\alpha(X) \otimes \Phi_\phi^{\beta\delta}(Y, Z))], \quad (94)$$

where the sum over  $\gamma$  is implicitly such that  $|\gamma| = |\alpha| + |\beta|$  and the sum over  $\phi$  is implicitly such that  $|\phi| = |\beta| + |\delta|$ . Although  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$  is not necessarily real or positive, the above expression makes it evident that

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \geq 0, \quad \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \geq 0. \quad (95)$$

Below we summarize all of the summation identities satisfied by  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ :

$$\begin{array}{ccccccc} & & \xrightarrow{\sum_\gamma} & \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\sum_\epsilon} & \varphi_\alpha(X) \varphi_\phi^{\beta\delta}(Y, Z) & \xrightarrow{\sum_{\beta\delta}} \varphi_\alpha(X) \varphi_\phi(Y + Z) \\ & & & \downarrow \sum_\phi & & \downarrow \sum_\phi & \\ \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\sum_{\alpha\beta\delta\gamma\phi}} \varphi_\epsilon(X + Y + Z) & \xleftarrow{\sum_{\alpha\beta\delta}} \varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\sum_\epsilon} & \varphi_\alpha(X) \varphi_\beta(Y) \varphi_\delta(Z) & & \\ & & \uparrow \sum_\gamma & & \uparrow \sum_\gamma & & \\ & & \xrightarrow{\sum_\phi} & \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\sum_\epsilon} & \varphi_\gamma^{\alpha\beta}(X, Y) \varphi_\delta(Z) & \xrightarrow{\sum_{\alpha\beta}} \varphi_\gamma(X + Y) \varphi_\delta(Z) \end{array}$$

The next series of results concerns upper bounds on the absolute value of  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ . These upper bounds follow from the following general lemma applied to the compact expression for  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$  appearing in Eq. (92).

*lem:cs\_ineq\_double\_proj*

**Lemma 42.** *Let  $\Omega \geq 0$  be a positive semidefinite operator and let  $\Pi_1, \Pi_2$  be Hermitian projection operators acting on a common vector space. Then*

$$|\text{Tr}(\Pi_1 \Pi_2 \Omega)| \leq \|\Pi_1 \Pi_2\|_\infty \text{Tr}(\Pi_1 \Omega)^{\frac{1}{2}} \text{Tr}(\Pi_2 \Omega)^{\frac{1}{2}}. \quad (96)$$

*Proof.* As  $\Omega \geq 0$  is positive semidefinite, let  $\Omega^{\frac{1}{2}} \geq 0$  denote its square root. Then let  $L = \Pi_1 \Pi_2 \Omega^{\frac{1}{2}}$  and  $R = \Pi_1 \Omega^{\frac{1}{2}}$  so that

$$\langle L, R \rangle = \text{Tr}(L^\dagger R) = \text{Tr}(\Pi_1 \Pi_2 \Omega \Pi_1) = \text{Tr}(\Pi_1 \Pi_2 \Omega), \quad (97)$$

where the last equality uses the assumption that  $\Pi_1^2 = \Pi_1$ . An application of the Cauchy-Schwarz inequality for the Hilbert-Schmidt inner product  $\langle L, R \rangle$  then implies

$$|\text{Tr}(\Pi_1 \Pi_2 \Omega)|^2 = |\langle L, R \rangle|^2 \leq \text{Tr}[L^\dagger L] \text{Tr}[R^\dagger R] = \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1] \text{Tr}[\Pi_1 \Omega \Pi_1] = \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2] \text{Tr}[\Pi_1 \Omega]. \quad (98)$$

Now let  $\{v_j\}_j$  be an orthonormal eigenbasis for  $\Pi_2 \Omega \Pi_2$  with eigenvalue  $\omega_j \geq 0$ . Then, because it was assumed that  $\Pi_2^2 = \Pi_2$ , we have

$$\text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2] = \text{Tr}[(\Pi_2 \Pi_1^2 \Pi_2)(\Pi_2 \Omega \Pi_2)] = \sum_j \omega_j \langle \Pi_1 \Pi_2 v_j, \Pi_1 \Pi_2 v_j \rangle \leq \sum_j \omega_j \|\Pi_1 \Pi_2\|_\infty^2 = \text{Tr}[\Pi_2 \Omega] \|\Pi_1 \Pi_2\|_\infty^2. \quad (99)$$

The claim then follows from combining these inequalities and taking the square root of both sides.  $\square$

**Theorem 43.** Let  $X, Y, Z \geq 0$  be positive semidefinite  $n \times n$  matrices. Then for all sextuples of Young diagrams  $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ ,

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)^{\frac{1}{2}} \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)^{\frac{1}{2}}. \quad (100)$$

*Proof.* Under the assumption that  $X, Y, Z \geq 0$ , we have that  $\Omega_{\alpha\beta\delta}^{X,Y,Z} = \Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z) \geq 0$  is positive semidefinite also. Now applying Lemma 42 to the compact expression for  $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$  in Eq. (92) yields

$$\left| \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}] \right| \leq \left\| \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \right\|_{\infty} \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Omega_{\alpha\beta\delta}^{X,Y,Z}]^{\frac{1}{2}} \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}]^{\frac{1}{2}} \quad (101)$$

which is equivalent to the claim due to Proposition 40.  $\square$

cor:tet\_eigenvalue\_min

**Corollary 44.** Let  $X, Y, Z \geq 0$  be positive semidefinite  $n \times n$  matrices. Then for all triples of Young digrams  $(\alpha, \beta, \delta)$ ,

$$\varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z) \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty} \min\{(\varphi_{\alpha}(X) \varphi_{\phi}(Y + Z) \varphi_{\gamma}(X + Y) \varphi_{\delta}(Z))^{\frac{1}{2}}, \varphi_{\epsilon}(X + Y + Z)\} \quad (102)$$

*Proof.* The claim follows by summing the inequality in Theorem 43 over triples  $(\gamma, \epsilon, \phi)$  and relaxing both the sides of the resulting inequality. For the lower bound, we have  $\varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z) = \sum_{\gamma\epsilon\phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$  which implies

$$\varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z) \leq \sum_{\gamma\epsilon\phi} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|. \quad (103)$$

Now for the upper bound, we have (i)  $\varphi_{\alpha}(X) \varphi_{\phi}(Y + Z) = \sum_{\epsilon\beta\delta} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$  which implies

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_{\alpha}(X) \varphi_{\phi}(Y + Z), \quad (104)$$

(ii)  $\varphi_{\gamma}(X + Y) \varphi_{\delta}(Z) = \sum_{\epsilon\alpha\beta} \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$  which implies

$$\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_{\gamma}(X + Y) \varphi_{\delta}(Z), \quad (105)$$

and (iii)  $\varphi_{\epsilon}(X + Y + Z) = \sum_{\alpha\beta\delta} \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z)$  which implies

$$\max\{\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z), \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)\} \leq \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_{\epsilon}(X + Y + Z), \quad (106)$$

Combining (i), (ii) and (iii),

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \leq \min\{\varphi_{\alpha}(X) \varphi_{\phi}(Y + Z) \varphi_{\gamma}(X + Y) \varphi_{\delta}(Z), \varphi_{\epsilon}(X + Y + Z)^2\}, \quad (107)$$

which then yields the claim.  $\square$

cor:tet\_eigenvalue\_cubic

**Corollary 45.** Let  $X, Y, Z \geq 0$  be positive semidefinite  $n \times n$  matrices. Then for all triples of Young digrams  $(\alpha, \beta, \delta)$ ,

$$\varphi_{\alpha}(X)^{\frac{2}{3}} \varphi_{\beta}(Y) \varphi_{\delta}(Z)^{\frac{2}{3}} \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty} \varphi_{\phi}(Y + Z)^{\frac{1}{3}} \varphi_{\gamma}(X + Y)^{\frac{1}{3}} \varphi_{\epsilon}(X + Y + Z)^{\frac{1}{3}}. \quad (108)$$

*Proof.* The proof is analogous to Corollary 45, except where we supplant Eq. (107) with the inequality

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \leq [\varphi_{\alpha}(X) \varphi_{\phi}(Y + Z) \varphi_{\gamma}(X + Y) \varphi_{\delta}(Z)]^p \varphi_{\epsilon}(X + Y + Z)^{2(1-p)}, \quad (109)$$

which holds for any  $p \in [0, 1]$ . The claim corresponds to the choice  $p = \frac{2}{3}$  which makes the exponents match.  $\square$

### 3.7 Eigenvalue estimation & separation

**Definition 46.** The Schur-Weyl distribution of order  $k$  for a positive semidefinite matrix  $X \geq 0$  with eigenvalues  $x = (x_1, \dots, x_n)$  is defined as the probability distribution over Young diagrams  $\lambda$  with  $k$  boxes with probability

$$p_X(\lambda) := \frac{\text{Tr}[P_\lambda X^{\otimes k}]}{\text{Tr}[X^{\otimes k}]} = \frac{\varphi_\lambda(X)}{\sum_{\lambda \vdash k} \varphi_\lambda(X)} = \frac{\dim W_\lambda s_\lambda(x)}{|x|^k} = \frac{k!}{|x|^k} \varphi_\lambda(X). \quad (110)$$

**Proposition 47.** Given a Young diagram  $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$ , and non-negative eigenvalues  $x = (x_1 \geq \dots \geq x_n \geq 0)$ ,

$$x_1^{\lambda_1} x_2^{\lambda_2} \dots x_n^{\lambda_n} \leq s_\lambda(x) \leq \dim V_\lambda^n x_1^{\lambda_1} x_2^{\lambda_2} \dots x_n^{\lambda_n}, \quad H_\lambda \geq \lambda_1! \dots \lambda_n!, \quad (111)$$

where  $\dim V_\lambda^n = s_\lambda(1, \dots, 1)$  counts the number of semistandard Young tableaux and therefore

$$\frac{k!}{|x|^k} \varphi_\lambda(x) = p_X(\lambda) \leq \dim V_\lambda^n m_{\underline{x}}(\lambda) \quad (112)$$

where  $m_{\underline{x}}(\lambda)$  is the multinomial distribution

$$m_{\underline{x}}(\lambda) := \frac{k!}{|x|^k} \frac{x_1^{\lambda_1}}{\lambda_1!} \frac{x_2^{\lambda_2}}{\lambda_2!} \dots \frac{x_n^{\lambda_n}}{\lambda_n!}. \quad (113)$$

**Remark 48.** Throughout this paper, we make frequent use of the dimension bound

$$\dim V_\lambda^n \leq m_n(k) := (k+1)^{\frac{n(n-1)}{2}}. \quad (114)$$

**Proposition 49** (Eigenvalue estimation). Let  $X \geq 0$  be a positive semidefinite  $n \times n$  matrix with eigenvalues  $x = (x_1 \geq \dots \geq x_n \geq 0)$ . Then the Schur-Weyl distribution  $p_X(\lambda)$  for  $\lambda \vdash k$  satisfies

$$(k+n)^{-\frac{(n+2)(n-1)}{2}} \exp(-kD(\lambda \parallel \underline{x})) \leq p_X(\lambda) \leq (k+1)^{\frac{n(n-1)}{2}} \exp(-kD(\lambda \parallel \underline{x})) \quad (115)$$

where  $D(\lambda \parallel \underline{x})$  denotes the Kullback-Leibler divergence of the normalized Young diagram  $\lambda$  from the normalized eigenvalues  $\underline{x}$ .

**Proposition 50** (Eigenvalue separation). Let  $(x, y)$  be eigenvalues of positive semidefinite  $n \times n$  matrices  $X, Y \geq 0$ . Then the fidelity  $F_k(p_X, p_Y) := \sum_{\lambda \vdash k} \sqrt{p_X(\lambda) p_Y(\lambda)}$  between Schur-Weyl distributions satisfies

$$F_k(p_X, p_Y) \leq (k+1)^{n(n-1)/2} F_k(m_{\underline{x}}, m_{\underline{y}}) = (k+1)^{n(n-1)/2} F(\underline{x}, \underline{y})^k. \quad (116)$$

**Proposition 51** (Eigenvalue separation). Let  $X, X' \geq 0$  be positive semidefinite  $n \times n$  matrices with the same trace  $\tau_X = \text{Tr}(X) = \text{Tr}(X')$ . Then for any integer  $k \geq 0$ ,

$$\sum_{\lambda \vdash k} \sqrt{\varphi_\lambda(X) \varphi_\lambda(X')} \leq (k+1)^{n(n-1)/2} \frac{\tilde{F}(x, x')^k}{k!}. \quad (117)$$

where  $\tilde{F}(x, x') := \sum_{i=1}^n \sqrt{x_i x'_i}$  denotes the unnormalized fidelity between  $x$  and  $x'$  (it is strictly less than  $\tau_X$  when  $x \neq x'$  and equals  $\tau_X$  otherwise).

**Corollary 52** (Multi-eigenvalue separation). Let  $X, Y \geq 0$  and  $X', Y' \geq 0$  be two pairs of  $n \times n$  matrices with matching traces  $\tau_X = \text{Tr}(X) = \text{Tr}(X')$  and  $\tau_Y = \text{Tr}(Y) = \text{Tr}(Y')$ . Then for any integer  $k \geq 0$ ,

$$\sum_{\ell=0}^k \sum_{\alpha \vdash \ell, \beta \vdash k-\ell} \sqrt{\varphi_\alpha(X) \varphi_\alpha(X')} \sqrt{\varphi_\beta(Y) \varphi_\beta(Y')} \leq (k^2/4 + k + 1)^{n(n-1)/2} \frac{\tilde{F}(X \oplus Y, X' \oplus Y')^k}{k!}. \quad (118)$$

*Proof.* The proof uses  $\tilde{F}(X \oplus Y, X' \oplus Y') = \tilde{F}(X, X') + \tilde{F}(Y, Y')$  and the binomial expansion. The prefactor arises because if  $\alpha \vdash \ell$  and  $\beta \vdash k - \ell$ , then

$$\dim(V_\alpha^n) \dim(V_\beta^n) \leq (\ell+1)^{n(n-1)/2} (k-\ell+1)^{n(n-1)/2} = (\ell(k-\ell) + k+1)^{n(n-1)/2}, \quad (119)$$

and  $\ell(k-\ell) \leq (\frac{k}{2})^2$ . □

**Remark 53.** The fidelity  $F(p, q)$  between distributiosn  $p, q$  is related to the total variation distance  $TV(p, q) = \frac{1}{2} \|p - q\|_1$  via the inequalities:

$$1 - TV(p, q) \leq F(p, q) \leq \sqrt{1 - TV(p, q)^2}. \quad (120)$$

Alternatively, the Bhattacharyya coefficient  $B(p, q) = -\log F(p, q)$  satisfies  $B(p, q) \geq \frac{1}{2} TV(p, q)^2 = \frac{1}{4} \|p - q\|_1^2$  which implies

$$F(p, q)^k = \exp(-kB(p, q)) \leq \exp\left(-\frac{1}{4} \|p - q\|_1^2\right). \quad (121)$$

### 3.8 Proof of Theorem 4

To begin we use the random matrix identities from Section 3.3 to prove that  $\varphi_\gamma(A + B)$  does not differ significantly when  $A, B$  are replaced with random matrices in their respective coadjoint orbits. This result follows from Corollary 38.

*cor:random\_sum\_ineq*

**Corollary 54.** For a given  $\gamma$ , let  $m_n(\gamma)$  be the following quantity

$$m_n(\gamma) := \min_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \frac{c_{\alpha\beta}^\gamma \dim V_\gamma^n}{\dim V_\alpha^n \otimes V_\beta^n} \quad (122)$$

Then for all  $A, B \geq 0$ ,

$$\varphi_\gamma(A + B) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\alpha(A) \varphi_\beta(B) \leq \frac{1}{m_n(\gamma)} \mathbb{E}_{A' \sim \mathcal{O}_a} \mathbb{E}_{B' \sim \mathcal{O}_b} \varphi_\gamma(A' + B') \quad (123)$$

Before continuing to the proof of Theorem 4, we remark that the prefactor  $\frac{c_{\alpha\beta}^\gamma \dim V_\gamma^n}{\dim V_\alpha^n \otimes V_\beta^n}$  above can be viewed as a probability distribution over Young diagrams  $\gamma$  of size  $|\gamma| = |\alpha| + |\beta|$  when conditioned on fixed  $\alpha, \beta$  because it corresponds to the proportion of  $V_\alpha^n \otimes V_\beta^n$  occupied by the  $\gamma$ -isotypic subspace:

$$\sum_{\gamma} c_{\alpha\beta}^\gamma \dim V_\gamma^n = \dim(V_\alpha^n \otimes V_\beta^n). \quad (124)$$

From this point of view, the quantity  $m_n(\gamma)$  can be interpreted as the minimum likelihood for the outcome  $\gamma$  as one varies over all pairs of Young diagrams  $(\alpha, \beta)$  of sizes fixed by  $\gamma$ .

*Proof of Theorem 4.* That Eq. (3) must be satisfied by  $(a, b, c)$  whenever there exists follows from Corollary 38. To prove the bound on  $D(a, b, c)$ , it actually suffices for  $(a, b, c)$  to satisfy a weaker constraint. Specifically, the requirement that Eq. (3) is satisfied for a given  $\gamma$  can be relaxed, by Corollary 54, to the requirement that

$$\varphi_\gamma(c) \leq \frac{1}{m_n(\gamma)} \max_{(X, Y) \in \mathcal{O}_a \times \mathcal{O}_b} \varphi_\gamma(X + Y). \quad (125)$$

Since it was assumed that  $|c| = |a| + |b| = \text{Tr}(X)$ , an application of the eigenvalue separation lemma produces:

$$\varphi_\gamma(c) = \sqrt{\varphi_\gamma(c)} \sqrt{\varphi_\gamma(c)}, \quad (126)$$

$$\leq \sqrt{\frac{1}{m_n(\gamma)} \max_{(X, Y) \in \mathcal{O}_a \times \mathcal{O}_b} \varphi_\gamma(c) \varphi_\gamma(X + Y)}, \quad (127)$$

$$\leq \frac{|c|^k}{k!} \sqrt{\frac{(\dim V_\gamma^n)^2}{m_n(\gamma)}} \exp\left(-\frac{k}{4|c|^2} D(a, b, c)^2\right). \quad (128)$$

*{eq:eig\_sep\_lr}*

Furthermore, note that<sup>7</sup>

$$\frac{(\dim V_\gamma^n)^2}{m_n(\gamma)} = \max_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \frac{\dim V_\alpha^n \otimes V_\beta^n \otimes V_\gamma^n}{c_{\alpha\beta}^\gamma} \leq \left( (k+1)^{n(n-1)/2} \right)^3. \quad (129)$$

Since this constraint is assumed to hold for all  $\gamma \vdash k$ , we may sum over all  $\gamma \vdash k$ . After summing and dividing both sides of Eq. (128) by  $\frac{|c|^k}{k!}$  (and noting there are at most  $(k+1)^{n-1}$  summands) yields

$$1 \leq (k+1)^{3n(n-1)/4+n-1} \exp \left( -\frac{k}{4|c|^2} D(a, b, c)^2 \right). \quad (130)$$

Rearranging for  $D(a, b, c)$  then yields the claimed bound:

$$\frac{1}{|c|} D(a, b, c) \leq \sqrt{(3n^2 + n - 4) \frac{\log(k+1)}{k}}. \quad (131)$$

□

### 3.9 Proof of Theorem 2

*Proof of Theorem 2.* If a sextuple of eigenvalues  $(a, b, c, d, e, f)$  happens to satisfy Eq. (1) for a particular triple of Young diagrams  $(\alpha, \beta, \delta)$ , then multiplying both sides of Eq. (1) by  $\varphi_\alpha(a)^{\frac{4}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{4}{3}}$  yields

$$(\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d))^2 \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_\alpha(a)^{\frac{4}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{4}{3}} \varphi_\phi(f)^{\frac{1}{3}} \varphi_\gamma(c)^{\frac{1}{3}} \varphi_\epsilon(e)^{\frac{1}{3}}. \quad (132)$$

Our objective is to sum this expression over all triples  $(\alpha, \beta, \delta)$  such that  $|\alpha| + |\beta| + |\delta| = k$ . Note that summing over  $\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d)$  yields

$$\sum_{\substack{\alpha, \beta, \delta \\ |\alpha| + |\beta| + |\delta| = k}} \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) = \frac{|e|^k}{k!} \quad (133)$$

since we explicitly assumed the trace conditions held, i.e.,  $|e| = |a| + |b| + |d|$ .

For notational convenience, let  $W_k \left( \begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right)$  denote the corresponding sum of the right-hand side (appropriately normalized by  $\frac{|e|^k}{k!}$  so that  $W_k$  is invariant under rescaling):

$$W_k \left( \begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right) := \left( \frac{k!}{|e|^k} \right)^2 \sum_{\substack{\alpha, \beta, \delta \\ \gamma, \epsilon, \phi \\ |\epsilon| = k}} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_\alpha(a)^{\frac{4}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{4}{3}} \varphi_\phi(f)^{\frac{1}{3}} \varphi_\gamma(c)^{\frac{1}{3}} \varphi_\epsilon(e)^{\frac{1}{3}}. \quad (134)$$

Note that  $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty$  is only non-zero if the size of the Young diagrams satisfy the size conditions,

$$k = |\epsilon| = |\alpha| + |\beta| + |\delta| = |\alpha| + |\phi| = |\gamma| + |\delta|, \quad (135)$$

and each of  $\varphi_\alpha(a), \varphi_\beta(b), \varphi_\delta(d), \varphi_\phi(f), \varphi_\gamma(c)$  and  $\varphi_\epsilon(e)$  are respectively only non-zero if the number of rows in the Young diagrams satisfy the rank constraints:

$$\ell(\alpha) \leq \text{rank}(a), \quad \ell(\beta) \leq \text{rank}(b), \quad \ell(\gamma) \leq \text{rank}(c), \quad \ell(\delta) \leq \text{rank}(d), \quad \ell(\epsilon) \leq \text{rank}(e), \quad \ell(\phi) \leq \text{rank}(f). \quad (136)$$

---

<sup>7</sup>TC: bound can be improved a bit by noting that  $k = |\gamma| = |\alpha| + |\gamma|$  implies  $(|\alpha| + 1)(|\beta| + 1) = |\alpha||\beta| + |\gamma| + 1 \leq \frac{k^2}{4} + k + 1$ .

The bound in Eq. (132) then implies a moment inequality bound:

$$W_k \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \geq \frac{1}{(k+1)^{3(n-1)}}, \quad (137)$$

where  $(k+1)^{3(n-1)}$  is an upper bound on the number of triples  $(\alpha, \beta, \delta)$  such that  $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \neq 0$  as there are at most  $(k+1)^{\ell-1}$  Young diagrams with  $k$  boxes and  $\ell$  rows.

Our objective now is to derive a complementary upper bound on  $W_k \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$ . To accomplish this, we note that an immediate corollary of Lemma 36 is the following bound which holds for any triple  $(x, y, z)$  of positive semidefinite eigenvalues:

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \varphi_\alpha(x)\varphi_\beta(y)\varphi_\delta(z) \leq R_\epsilon^{\alpha\beta\delta} \max_{\substack{(X,Y,Z) \\ \in \mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}_z}} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|, \quad (138)$$

where  $R_\epsilon^{\alpha\beta\delta} = \frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)}$  and the where the maximum is taken over all triples  $(X, Y, Z)$  of positive semidefinite matrices with eigenvalues  $(x, y, z)$ . Applying Theorem 43 (and its related corollaries) to  $\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|$  on the right-hand side above produces

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty \varphi_\phi(Y+Z)^{\frac{1}{3}} \varphi_\gamma(X+Y)^{\frac{1}{3}} \varphi_\epsilon(X+Y+Z)^{\frac{1}{3}}. \quad \text{\textcolor{blue}{eq:sup_bound}} \quad (139)$$

Then Eq. (139) can be applied to each summand in  $W_k \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$  (replacing  $(x, y, z)$  with  $(a, b, d)$ ) to obtain

$$W_k \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq \left( \frac{k!}{|e|^k} \right)^2 \sum_{\substack{\alpha, \beta, \delta \\ \gamma, \epsilon, \phi \\ |\epsilon|=k}} R_\epsilon^{\alpha\beta\delta} \varphi_\alpha(a)^{\frac{2}{3}} \varphi_\delta(d)^{\frac{2}{3}} \max_{\substack{(X,Y,Z) \\ \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d}} [\varphi_\phi(f)\varphi_\phi(Y+Z)\varphi_\gamma(c)\varphi_\gamma(X+Y)\varphi_\epsilon(e)\varphi_\epsilon(X+Y+Z)]^{\frac{1}{3}}. \quad (140)$$

Note that here we have made use of the inequality  $\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \geq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty^2$  to simplify the presentation slightly. Now the eigenvalue separation lemma implies a triple of bounds (because  $|\alpha|+|\phi| = |\gamma|+|\delta| = |\epsilon| = k$ )

$$\varphi_\alpha(a) \sqrt{\varphi_\phi(f)\varphi_\phi(Y+Z)} \leq \dim(V_\alpha^n \otimes V_\phi^n) \frac{|e|^k}{k!} \exp \left( -\frac{k}{4|e|^2} \|f - \text{eig}(Y+Z)\|_1^2 \right), \quad (141)$$

$$\varphi_\delta(d) \sqrt{\varphi_\gamma(c)\varphi_\gamma(X+Y)} \leq \dim(V_\gamma^n \otimes V_\delta^n) \frac{|e|^k}{k!} \exp \left( -\frac{k}{4|e|^2} \|c - \text{eig}(X+Y)\|_1^2 \right), \quad (142)$$

$$\sqrt{\varphi_\epsilon(e)\varphi_\epsilon(X+Y+Z)} \leq \dim(V_\epsilon^n) \frac{|e|^k}{k!} \exp \left( -\frac{k}{4|e|^2} \|e - \text{eig}(X+Y+Z)\|_1^2 \right). \quad (143)$$

Taking the product of these three inequalities to the power of  $2/3$ , applying it to the upper bound on  $W_k$  and combining it with the lower bound on  $W_k$  then implies

$$\min_{\substack{(X,Y,Z) \\ \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d}} \{ \|f - \text{eig}(Y+Z)\|_1^2 + \|c - \text{eig}(X+Y)\|_1^2 + \|e - \text{eig}(X+Y+Z)\|_1^2 \} \leq \frac{6|e|^2}{k} (3n(n-1) + 9(n-1)) \log(k+1). \quad (144)$$

where the  $6(n-1)$  factor is an upper bound on the number of non-zero summands for the right-hand side, and additional factor of  $3(n-1)$  comes from the lower bound on  $W_k$  and where the  $3n(n-1)$  factor comes from an upper bound to the dimensions:

$$\frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)} \dim(V_\alpha^n \otimes V_\phi^n \otimes V_\gamma^n \otimes V_\delta^n \otimes V_\epsilon^n)^{\frac{2}{3}} \leq \left[ (k+1)^{\frac{n(n-1)}{2}} \right]^{3+4\frac{2}{3}} \leq (k+1)^{3n(n-1)}. \quad (145)$$

Therefore, the deviation measure is bounded by

$$D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq |e| \sqrt{(18n^2 + 36n - 54) \frac{\log(k+1)}{k}} \quad (146)$$

which proves the claim.  $\square$

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## References

- [AB68] Vishnu K Agrawala and Johan G Belinfante. “Graphical formulation of recoupling theory for any compact group”. In: *Annals of Physics* 49.1 (1968), pp. 130–170.
- [AH11] Daiki Akimoto and Masahito Hayashi. “Discrimination of the change point in a quantum setting”. In: *Physical Review A* 83.5 (2011), p. 052328.
- [AI19] Arseniy Akopyan and Ivan Izmistiev. “The Regge symmetry, confocal conics, and the Schläfli formula”. In: *Bulletin of the London Mathematical Society* 51.5 (2019), pp. 765–775.
- [Alc+23] Judith Alcock-Zeilinger et al. “Wigner 6j symbols for SU (N): Symbols with at least two quark-lines”. In: *Journal of Mathematical Physics* 64.2 (2023).
- [Ale+25] Anton Alekseev et al. “Multiple Horn problems for planar networks and invertible matrices”. In: *arXiv preprint arXiv:2503.05277* (2025).
- [AMS20] Victor Alekseev, Andrey Morozov, and Alexey Sleptsov. “Multiplicity-free  $U_q(\mathfrak{sl}N)$  6-j symbols: relations, asymptotics, symmetries”. In: *Nuclear Physics B* 960 (2020), p. 115164.
- [AMW01] Anton Alekseev, Eckhard Meinrenken, and Chris Woodward. “Linearization of Poisson actions and singular values of matrix products”. In: *Annales de l’institut Fourier*. Vol. 51. 6. 2001, pp. 1691–1717.
- [Aqu+12] Vincenzo Aquilanti et al. “Semiclassical mechanics of the Wigner 6j-symbol”. In: *Journal of Physics A: Mathematical and Theoretical* 45.6 (2012), p. 065209.
- [ARS88] Robert Alicki, Sławomir Rudnicki, and Sławomir Sadowski. “Symmetry properties of product states for the system of N n-level atoms”. In: *Journal of mathematical physics* 29.5 (1988), pp. 1158–1162.
- [Art23] DV Artamonov. “Classical 6j-symbols of finite-dimensional representations of the algebra  $\mathfrak{gl}_3$ ”. In: *Theoretical and Mathematical Physics* 216.1 (2023), pp. 909–923.
- [Art24] Dmitry Artamonov. “Calculation of 6j-symbols for the Lie algebra  $\mathfrak{gl}_n$ ”. In: *arXiv preprint arXiv:2405.05628* (2024).
- [AW97] Sharad Agnihotri and Chris Woodward. “Eigenvalues of products of unitary matrices and quantum Schubert calculus”. In: *arXiv preprint alg-geom/9712013* (1997).
- [Bac10] Miriam Backens. “The 6j symbols and an extended version of Horn’s problem”. In: (2010).
- [Bae96] John C Baez. “Spin networks in gauge theory”. In: *Advances in Mathematics* 117.2 (1996), pp. 253–272.
- [Bae99] John C Baez. “An introduction to spin foam models of quantum gravity and BF theory”. In: (1999). arXiv: [gr-qc/9905087](https://arxiv.org/abs/gr-qc/9905087).
- [Bar03] John W Barrett. “Geometrical measurements in three-dimensional quantum gravity”. In: *International Journal of Modern Physics A* 18.supp02 (2003), pp. 97–113.
- [BC98] John W Barrett and Louis Crane. “Relativistic spin networks and quantum gravity”. In: *Journal of Mathematical Physics* 39.6 (1998), pp. 3296–3302.
- [Bel02] Prakash Belkale. “Geometric proofs of Horn and saturation conjectures”. In: *arXiv preprint math/0208107* (2002).
- [Bel05] Prakash Belkale. “Geometric proof of a conjecture of Fulton”. In: *arXiv preprint math/0511664* (2005).
- [Bel08] Prakash Belkale. “Quantum generalization of the Horn conjecture”. In: *Journal of the American Mathematical Society* 21.2 (2008), pp. 365–408.
- [Bha01] Rajendra Bhatia. “Linear algebra to quantum cohomology: the story of Alfred Horn’s inequalities”. In: *The American Mathematical Monthly* 108.4 (2001), pp. 289–318.
- [BK16] Prakash Belkale and Shrawan Kumar. “The multiplicative eigenvalue problem and deformed quantum cohomology”. In: *Advances in Mathematics* 288 (2016), pp. 1309–1359.

- [BN23] Bruce Bartlett and Nzaganya Nzaganya. “Coherent loop states and angular momentum”. In: *arXiv preprint arXiv:2306.17293* (2023).
- [Boa07] Philip P Boalch. “Regge and Okamoto symmetries”. In: *Communications in mathematical physics* 276.1 (2007), pp. 117–130.
- [BS10] John Baez and Mike Stay. “Physics, topology, logic and computation: a Rosetta Stone”. In: *New structures for physics*. Springer, 2010, pp. 95–172. arXiv: [0903.0340v3](#).
- [BVW18] Nicole Berline, Michèle Vergne, and Michael Walter. “The Horn inequalities from a geometric point of view”. In: *L’Enseignement Mathématique* 63.3 (2018), pp. 403–470.
- [Chr08] Matthias Christandl. “A Quantum Information-Theoretic Proof of the Relation between Horn’s Problem and the Littlewood-Richardson Coefficients”. In: *Lecture Notes in Computer Science*. Springer Berlin Heidelberg, 2008, pp. 120–128.
- [CHW07] Andrew M Childs, Aram W Harrow, and Paweł Wocejn. “Weak Fourier-Schur sampling, the hidden subgroup problem, and the quantum collision problem”. In: *Annual Symposium on Theoretical Aspects of Computer Science*. Springer. 2007, pp. 598–609.
- [CIS99] Li-Chen Chen, Mourad EH Ismail, and Plamen Simeonov. “Asymptotics of Racah coefficients and polynomials”. In: *Journal of Physics A: Mathematical and General* 32.3 (1999), p. 537.
- [CMZ19] Robert Coquereaux, Colin McSwiggen, and Jean-Bernard Zuber. “Revisiting Horn’s problem”. In: *Journal of Statistical Mechanics: Theory and Experiment* 2019.9 (2019), p. 094018.
- [Coq23] Robert Coquereaux. “About integer-valued variants of the theta and 6j symbols”. In: *Journal of Mathematical Physics* 64.3 (2023).
- [CŞW18] Matthias Christandl, M Burak Şahinoğlu, and Michael Walter. “Recoupling coefficients and quantum entropies”. In: *Annales Henri Poincaré*. Vol. 19. 2. Springer. 2018, pp. 385–410.
- [CZ18] Robert Coquereaux and Jean-Bernard Zuber. “From orbital measures to Littlewood–Richardson coefficients and hive polytopes”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 339–386.
- [DH05] Sumit Daftuar and Patrick Hayden. “Quantum state transformations and the Schubert calculus”. In: *Annals of Physics* 315.1 (2005), pp. 80–122.
- [DJN93] Bergfinnur Durhuus, Hans Plesner Jakobsen, and Ryszard Nest. “Topological quantum field theories from generalized 6j-symbols”. In: *Reviews in Mathematical Physics* 5.01 (1993), pp. 1–67.
- [Edm96] Alan Robert Edmonds. *Angular momentum in quantum mechanics*. Vol. 4. Princeton university press, 1996.
- [Fan49] Ky Fan. “On a theorem of Weyl concerning eigenvalues of linear transformations I”. In: *Proceedings of the National Academy of Sciences* 35.11 (1949), pp. 652–655.
- [FK00] Laurent Freidel and Kirill Krasnov. “Simple spin networks as Feynman graphs”. In: *Journal of Mathematical Physics* 41.4 (2000), pp. 1681–1690.
- [FKL10] Laurent Freidel, Kirill Krasnov, and Etera R Livine. “Holomorphic factorization for a quantum tetrahedron”. In: *Communications in Mathematical Physics* 297.1 (2010), pp. 45–93.
- [FL03] Laurent Freidel and David Louapre. “Asymptotics of 6j and 10j symbols”. In: *Classical and Quantum Gravity* 20.7 (2003), p. 1267.
- [Fra22] Thomas C Fraser. *A sufficient family of necessary inequalities for the compatibility of quantum marginals*. 2022. arXiv: [2211.00685v1](#).
- [Ful00] William Fulton. “Eigenvalues, invariant factors, highest weights, and Schubert calculus”. In: *Bulletin of the American Mathematical Society* 37.3 (2000), pp. 209–249.
- [Ful98] William Fulton. “Eigenvalues of sums of Hermitian matrices”. In: *Séminaire Bourbaki* 40.255-269 (1998), p. 5.
- [GS82] Victor Guillemin and Shlomo Sternberg. “Geometric quantization and multiplicities of group representations”. In: *Inventiones mathematicae* 67.3 (1982), pp. 515–538.

- [Gur08] Razvan Gurau. “The Ponzano–Regge Asymptotic of the 6 j Symbol: An Elementary Proof”. In: *Annales Henri Poincare*. Vol. 9. Springer. 2008, pp. 1413–1424.
- [GW00] Roe Goodman and Nolan R Wallach. *Representations and invariants of the classical groups*. Cambridge University Press, 2000.
- [Hal08] Matthew John Hall. “Representations of the Symmetric Groups and Combinatorics of the Frobenius-Young Correspondence”. PhD thesis. Citeseer, 2008.
- [Hay17] Masahito Hayashi. *A Group Theoretic Approach to Quantum Information*. Springer, 2017.
- [HK96] Jean-Claude Hausmann and Allen Knutson. “Polygon spaces and Grassmannians”. In: *arXiv preprint dg-ga/9602012* (1996).
- [HM02] Masahito Hayashi and Keiji Matsumoto. “Quantum universal variable-length source coding”. In: *Physical Review A* 66.2 (2002), p. 022311.
- [Hor62] Alfred Horn. “Eigenvalues of sums of Hermitian matrices.” In: (1962).
- [IN66] Claude Itzykson and Michael Nauenberg. “Unitary groups: Representations and decompositions”. In: *Reviews of Modern Physics* 38.1 (1966), p. 95.
- [IP23] Christian Ikenmeyer and Greta Panova. “All Kronecker coefficients are reduced Kronecker coefficients”. In: *arXiv preprint arXiv:2305.03003* (2023).
- [Kir04] Aleksandr Aleksandrovich Kirillov. *Lectures on the orbit method*. Vol. 64. American Mathematical Soc., 2004.
- [Kly00] Alexander A Klyachko. “Random walks on symmetric spaces and inequalities for matrix spectra”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 37–59.
- [Kly98] Alexander A Klyachko. “Stable bundles, representation theory and Hermitian operators”. In: *Selecta Mathematica* 4 (1998), pp. 419–445.
- [Knu00] Allen Knutson. “The symplectic and algebraic geometry of Horn’s problem”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 61–81.
- [KPS24] Stefan Keppeler, Simon Plätzer, and Malin Sjö Dahl. “Wigner 6j symbols with gluon lines: completing the set of 6j symbols required for color decomposition”. In: *Journal of High Energy Physics* 2024.5 (2024), pp. 1–25.
- [KT99] Allen Knutson and Terence Tao. “The honeycomb model of  $GL_n(\mathbb{C})$  tensor products I: Proof of the saturation conjecture”. In: *Journal of the American Mathematical Society* 12.4 (1999), pp. 1055–1090.
- [KW01] Michael Keyl and Reinhard F Werner. “Estimating the spectrum of a density operator”. In: *Physical Review A* 64.5 (2001), p. 052311.
- [Lid53] V B Lidskiĭ. “The proper values of the sum and product of symmetric matrices”. Trans. by Curtis D. Benster. In: (1953). Ed. by George E. Forsythe Wallace Givens.
- [Lit58] Dudley E Littlewood. “Products and plethysms of characters with orthogonal, symplectic and symmetric groups”. In: *Canadian Journal of Mathematics* 10 (1958), pp. 17–32.
- [LS23] Elena Lanina and Alexey Sleptsov. “Tug-the-hook symmetry for quantum 6j-symbols”. In: *Physics Letters B* 845 (2023), p. 138138.
- [MMS17] A Mironov, A Morozov, and A Sleptsov. “On 6 j-symbols for symmetric representations of  $U_q(\mathfrak{su}_N)$ ”. In: *JETP Letters* 106 (2017), pp. 630–636.
- [Mou84] J Moussouris. “Quantum models of space-time based on recoupling theory”. PhD thesis. University of Oxford, 1984.
- [MR05] Annalisa Marzuoli and Mario Rasetti. “Computing spin networks”. In: *Annals of Physics* 318.2 (2005), pp. 345–407.
- [MS18] Andrey Morozov and Alexey Sleptsov. “New Symmetries for the  $U_q(\mathfrak{sl}_N)$  6j Symbols from the Eigenvalue Conjecture”. In: *JETP Letters* 108 (2018), pp. 697–704.

- [Mur55] Francis D Murnaghan. “On the analysis of the Kronecker product of irreducible representations of  $S_n$ ”. In: *Proceedings of the National Academy of Sciences* 41.7 (1955), pp. 515–518.
- [Nis00] Kyo Nishiyama. “Restriction of the irreducible representations of  $GL_n$  to the symmetric group  $S_n$ ”. In: *preprint* (2000).
- [NP06] Constantin Niculescu and Lars-Erik Persson. *Convex functions and their applications*. Vol. 23. Springer, 2006.
- [NPZ13] Satoshi Nawata, Ramadevi Pichai, and Zodinmawia. “Multiplicity-free Quantum  $6j$ -Symbols for  $U_q(\mathfrak{sl}_N)$ ”. In: *Letters in Mathematical Physics* 103 (2013), pp. 1389–1398.
- [OW16] Ryan O’Donnell and John Wright. “Efficient quantum tomography”. In: *Proceedings of the forty-eighth annual ACM symposium on Theory of Computing*. 2016, pp. 899–912. arXiv: [1508.01907v2](#).
- [Pen71] Roger Penrose. “Applications of negative dimensional tensors”. In: *Combinatorial mathematics and its applications* 1 (1971), pp. 221–244.
- [PR68] Giorgio Ponzano and Tullio Eugenio Regge. “Semiclassical limit of Racah coefficients”. In: *Spectroscopic and group theoretical methods in physics* (1968).
- [Rac42] Giulio Racah. “Theory of complex spectra. II”. In: *Physical Review* 62.9-10 (1942), p. 438.
- [Reg59] Tullio Regge. “Symmetry properties of Racah’s coefficients”. In: *Il Nuovo Cimento (1955-1965)* 11.1 (1959), pp. 116–117.
- [Res87] N Yu Reshetikhin. *Quantized universal enveloping algebras, the Yang-Baxter equation and invariants of links, 1*. Tech. rep. 1987.
- [Ric02] G Richardson. “The trigonometry of the tetrahedron”. In: *The Mathematical Gazette* 2.32 (1902), pp. 149–158.
- [Rob02] Justin Roberts. “Asymptotics and  $6j$ -symbols”. In: *Geom. Topol. Monogr* 4 (2002), pp. 245–261.
- [Rob99] Justin Roberts. “[Classical  \$6j\$ -symbols and the tetrahedron](#)”. In: *Geometry & Topology* 3.1 (1999), pp. 21–66.
- [Rov04] Carlo Rovelli. *Quantum gravity*. Cambridge university press, 2004.
- [She15] Cass Sherman. “Geometric proof of a conjecture of King, Tollu, and Toumazet”. In: *arXiv preprint arXiv:1505.06551* (2015).
- [SJS18] Meera Sitharam, Audrey John, and Jessica Sidman, eds. *Handbook of geometric constraint systems principles*. Discrete Mathematics and Its Applications. New York, NY: Productivity Press, July 2018.
- [Spe05] David E Speyer. “Horn’s problem, Vinnikov curves, and the hive cone”. In: (2005).
- [TF70] Robert C. Thompson and Linda J. Freede. “[On the Eigenvalues of Sums of Hermitian Matrices. II.](#)” In: *Aequationes mathematicae* 5 (1970), pp. 103–115.
- [TF71] Robert C Thompson and Linda J Freede. “On the eigenvalues of sums of Hermitian matrices”. In: *Linear Algebra and Its Applications* 4.4 (1971), pp. 369–376.
- [TT71] RC Thompson and S Therianos. “The eigenvalues and singular values of matrix sums and product, viii: displacement of indices”. In: *aequationes mathematicae* 7 (1971), pp. 219–242.
- [TV92] Vladimir G Turaev and Oleg Ya Viro. “State sum invariants of 3-manifolds and quantum  $6j$ -symbols”. In: *Topology* 31.4 (1992), pp. 865–902.
- [TW06] Yuka U Taylor and Christopher T Woodward. “ $6j$  symbols for and non-Euclidean tetrahedra”. In: *Selecta Mathematica* 11.3-4 (2006), p. 539.
- [VMK88] Dmitrii Aleksandrovich Varshalovich, Anatolij Nikolaevič Moskalev, and Valerii Kel’manovich Khersonskii. [Quantum theory of angular momentum](#). World Scientific, 1988.
- [Wat99] James KG Watson. “Asymptotic approximations for certain  $6j$  and  $9j$  symbols”. In: *Journal of Physics A: Mathematical and General* 32.39 (1999), p. 6901.

- [WD09] Karl Wirth and André S Dreiding. “Edge lengths determining tetrahedrons”. In: *Elemente der Mathematik* 64.4 (2009), pp. 160–170.
- [Wey12] Hermann Weyl. “Das asymptotische Verteilungsgesetz der Eigenwerte linearer partieller Differentialgleichungen (mit einer Anwendung auf die Theorie der Hohlraumstrahlung)”. In: *Mathematische Annalen* 71.4 (1912), pp. 441–479.
- [Wie55] Helmut Wielandt. “An extremum property of sums of eigenvalues”. In: *Proceedings of the American Mathematical Society* 6.1 (1955), pp. 106–110.
- [Wig60] Eugene Paul Wigner. “Group theory and its application to the quantum mechanics of atomic spectra”. In: *American journal of physics* 28.4 (1960), pp. 408–409.
- [Wig93] Eugene P Wigner. “On the matrices which reduce the Kronecker products of representations of SR groups”. In: *The Collected Works of Eugene Paul Wigner: Part A: The Scientific Papers*. Springer, 1993, pp. 608–654.
- [Zub18] Jean-Bernard Zuber. “Horn’s problem and Harish-Chandra’s integrals. Probability density functions”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 309–338.